

# Which Escalation Signal Pays Depends on the Traffic: Free Baselines, Corpus Ceilings and Graded Actions in Visual Cascades Without Training

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[github.com/thekester/gwel](https://github.com/thekester/gwel)

## Abstract

Vision-language cascades answer from a downsampled image and escalate to full resolution when the model appears uncertain. Published policies learn that decision by reinforcement learning on 7B-class models and read it after generation. We ask what a sub-billion model supplies *before* it writes a token, and price every candidate against the baseline this literature omits: randomising between fixed configurations, which needs no signal and traces a convex hull any adaptive policy must clear.

The answer inverts between two regimes. Across a four-dataset mixture, output entropy never clears the hull on either serving model, since reading it requires completing the pass it prices; the image’s pixel dimensions, free from the file header, clear it everywhere, and the dataset label alone dominates every signal we test. Inside a single workload that descriptor is a weak ranker (0.508 to 0.586 AUROC on the escalation target across eight corpus-model pairs, against 0.609 to 0.791 for a model-read signal), yet its policy still clears on four of them, at a rate rising monotonically with how steeply the serving model prices resolution ( $\rho = 0.914$ ), and on one it clears where the model-read signal does not. **Which signal is worth reading is a property of the traffic, and a method evaluated only on a mixture will recommend the wrong one for a workload.**

We formalise what a hull gap certifies, namely information about cost and correctness *jointly* rather than about correctness, which is why a policy correlated with per-example cost escapes a mean-cost hull without reading value at all. Measured directly, that loophole reaches at most +0.008 on seven pairs, leaving our clearances there intact, and exceeds both signals on the eighth, which we withdraw.

What survives is graded action and measured cost. Thresholding the calibrated signed gain at the cost-implied break-even removes the escalation rate as a tunable; calibrating the weak pass’s error probability instead, which is cost-optimal under an assumption its authors verify for text and which visual escalation violates, over-spends on both models and escalates every query on the smaller. On 1200 document pages, 93% of what escalation repairs is repaired below the top rung, and accuracy saturates at a resolution four models agree on, at 640 to 1312 visual tokens. A second corpus does not share that ceiling: there a weak model’s null is its own capacity, which supplies

the precondition our procedure lacked. All comparisons are corrected for family-wise error over the two hundred and twenty-seven paired tests reported here, of which twenty-four do not survive.

## 1 Introduction

A vision-language model asked “what does the sign say?” can answer from a thumbnail or from the full-resolution image. The thumbnail costs a fraction as much and is often sufficient. Deciding *per query* which to use is the premise of adaptive visual perception, and the raw material for it is there: on our pilot, 34% of queries are answered correctly at 384px and would gain nothing from more pixels, 41% fail at both resolutions, and escalation changes the answer on 21%. Whether a per-query decision *pays*, against the simpler move of randomly assigning queries to fixed configurations at the same expected cost, is a separate question, and one this paper ends up answering more narrowly than it expected to.

Published systems make this decision by learning it and by reading it after generating: VisionThink [1], AdaptVision [2] and AwaRes [3] all train an escalation policy with reinforcement learning on 7B-class models, supervised by judge models of their own.

Below one billion parameters none of that is available. Curating twenty thousand samples and running GRPO is not an option for a model that exists because it fits in a gigabyte of GPU memory [9]. Our question is therefore narrow: *what does a sub-billion VLM already know about its own need for more pixels, and how cheaply can that be read?*

That scale is the constraint that makes the question sharp, not the object of study, and we are careful not to claim more. Every serving model that carries a result here has fewer than a billion parameters, and the central comparisons are measured on two of them. Nothing we measure shows the conclusions stopping at a billion: the two larger models we use as controls behave the same way on the saturation question, and whether the inversion of §5.1 or the cascade correction of §8 would survive at 7B is a question our budget does not reach.

Fig. 1 summarises the serving path: a single cheap pass, a score, and a choice among resolution rungs priced per example by the cost model of §3.2. The answer depends on the evaluation distribution. On a benchmark mixture

the model appears to know more than its output reveals; inside a single workload most of that apparent knowledge is a recognition of which benchmark the query came from. What survives is narrower: reading the right quantity matters more than reading it well, deciding *how far* to escalate matters more than deciding *whether*, and what the model knows is whether to look again rather than where.

## Contributions.

1. **An evaluation protocol for allocation signals, and the failure it detects.** Pooling examples across benchmarks that differ in how much resolution buys inflates any signal correlated with benchmark identity. Applied to a pre-generation probe, per-example cost accounting exposes it escalating 1.19 to 1.33 $\times$  larger images than entropy; scored inside a single dataset it reaches 0.572 where output entropy, fitted on nothing, reaches 0.663, at every depth and for four probe families; and the disagreement between the two signals predicts the source dataset at 0.738. The ranking advantage is a between-domain effect and does not survive family-wise correction (§7, §7.1).
2. **Which signal pays is a property of the traffic, and the two regimes require opposite answers.** Measured against randomisation between fixed configurations, which needs no signal and which the escalation literature does not use as a baseline: across a benchmark mixture, raw image size read from the file header clears the hull at every operating point and, on the 500M serving model, stays within 0.010 of a probe on activations, while output entropy never clears it on either sub-billion model tested; inside a workload the *ranking* inverts, image size falling to 0.508 AUROC against a model-read signal’s 0.731. The *policy* does not follow the ranking: the free descriptor still clears on four of eight corpus-model pairs, and how many preferences it clears at rises monotonically with how steeply the serving model prices escalation ( $\rho = 0.914$  over the eight). We measure what a policy reading only per-example price achieves against the same hull, which bounds the part of any gap that needs no signal: it is at most +0.008 on seven pairs, leaving our clearances there intact, and on the eighth it exceeds both signals and we withdraw that pair (§9). A policy that randomises with the probe’s mid-prefill abort clears nothing either, which places the probe’s margin in its signal and not in its read cost. A method evaluated only on a mixture will recommend the wrong signal for a workload (§5.1, §10).
3. **A decision rule derived from measured cost, and the cascade assumption it corrects.** Calibrating the expected signed gain and thresholding at the cost-implied break-even removes the rate parameter. Calibrating the weak model’s error probability instead, which is cost-optimal under the fixed-accuracy

condition [11] states and validates for text, coincides with our rule exactly when escalation cannot make things worse, and spends 59% relatively more compute here because it can (§8). The rule is a property of the method, not of any signal: run on output entropy inside each domain, one unchanged operator preference yields escalation rates from 1% to 61% tracking what escalation is worth there ( $r = 0.974$ ) (§8.1).

4. **How far to escalate, and where it stops paying.** Of the queries escalation repairs, 77% are repaired by an intermediate rung, and that rung is twice as efficient per millisecond as the one above it. On a single-domain pilot with rungs verified distinct per example, accuracy saturates at a pixel resolution that four models agree on, spanning 8.6 $\times$  parameters, two lineages and three ways of turning pixels into tokens while spending 640 to 1312 of them to get there, so a token budget cannot be what runs out. The ceiling is measurable once by a procedure we state and price at about 300 pages, and a second corpus supplies the precondition that procedure was missing: a null at the top step is a corpus property only when the serving model is not itself the binding constraint (§10, §7.1).
5. **A measured cost model.** Every comparison here is a cost comparison, so we measure rather than assume, and four standard proxies mispredict: the vision encoder is 43 to 50% of the visual cost, which caps post-encoder pruning at half; decode latency follows depth rather than parameter count; a flat per-configuration price understates escalation by 16%; and for 56% of the mixture the “full-resolution” configuration is not a distinct configuration at all (§3.2, §10).

**How to read this paper.** Three questions are asked in sequence and they receive different answers. §5 to §7.1 ask what a pre-generation signal is worth and conclude that on heterogeneous traffic it is worth what the image dimensions are worth, which is to say it need not be read from the model at all. §8 asks how a threshold should be set once a signal is chosen, and removes the rate parameter given a stated preference. §10 asks how far to escalate, and finds the answer to be a property of the corpus rather than of the query, measurable once by a stated procedure. Between these, §9 states what clearing the no-signal baseline certifies, measures how permissive that baseline is, and withdraws the one pair where it is too permissive to credit us. A reader interested only in deployment can read §10 and Algorithm 4 alone.

Beyond these, the paper reports several bounded results in their sections: the untrained self-report mechanism loses to uncertainty routing (§5.1); a localizer on pooled visual-token states beats random cell choice by 0.9 points, 6% of the gap a perfect one would close (§A); an ablation read at fixed budget rather than fixed preference (§B);

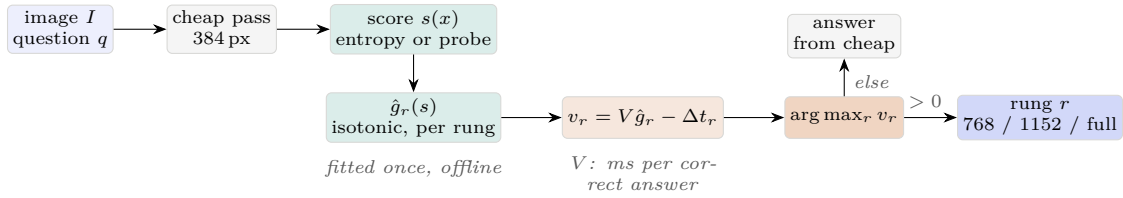


Figure 1: The serving path. Everything up to the score is a single cheap pass; the calibrators are fitted once offline and the operator supplies one number,  $V$ . Which rung is chosen, and whether any is, falls out of the measured latencies rather than a tuned rate (Algorithm 3).

and abstention priced on both of its axes (§E).

## 2 Related work

**Adaptive visual acquisition.** The three systems closest to this work all learn when to look closer. VisionThink [1] trains a special token requesting the full-resolution image with GRPO and an LLM-as-judge reward, plus a penalty calibrated to stop the policy collapsing into always escalating. AdaptVision [2] replaces the binary decision with a bounding-box tool call and introduces Decoupled Turn Policy Optimization to separate the credit assigned to tool tokens from that assigned to answer tokens. AwaRes [3] makes the coupled decision of *whether* and *where* jointly, with cold-start supervised fine-tuning followed by multi-turn GRPO, supervised automatically by a LLaMA-3.3-70B judge and an oracle grounding model. All three evaluate at 7B scale, all three learn the decision rather than reading it, and all three condition on the output of a completed pass. We differ on each count, and §5.1 measures what the first of those choices is worth by implementing their mechanism without its training.

**Visual token compression.** A parallel line reduces cost by discarding tokens after the vision encoder has run, typically ranking them by attention score. GlimpsePrune [4] is the variant we compare against in detail: it learns a data-driven pruning metric from a single-pass “glimpse” and removes 92.6% of visual tokens at parity. These methods and ours act on different halves of the same cost, which §3.2 quantifies: post-encoder pruning cannot recover the encoder’s share, and on the three models we measure that share is 43 to 50%.

**Routing from internal states.** In the text setting, several groups have shown that a model’s activations predict its own success before it generates. Lugoloobi et al. [5] train linear probes on pre-generation activations and route across a model pool at 40 to 70% lower cost; Moreno Cencerrado et al. [6] isolate a linear “in-advance correctness direction” that saturates at intermediate layers; NVIDIA’s LLM Router [7] predicts a target model’s correctness from a *different* model’s prefill activations, closing 45.6% of its oracle gap. Afzal et al. [19] report the

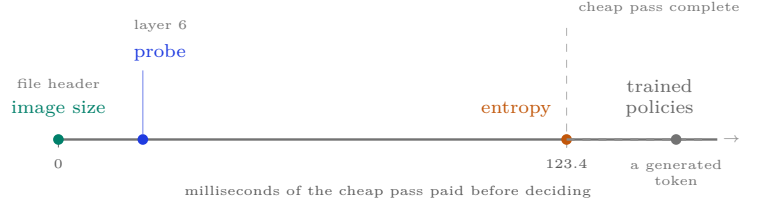


Figure 2: When each escalation decision becomes readable, on the cheap pass it is meant to price. The three published systems we compare against ([1, 2, 3], Table 1) emit the decision as a generated token, so the pass is complete and decoding has begun before anything can be read; output entropy needs the same complete pass; the probe is readable after a sixth of it; and raw image size needs no pass at all. **A signal that decides later is not merely reported later, it is strictly more expensive**, which is why §5.1 charges every candidate its read cost and why the cheapest signal on this axis is the hardest to beat on a mixture. Positions are measured for the first three; the trained policies are placed beyond the pass without a number, since we did not run their decode.

sharpest version of the premise we test: a probe on representations predicts whether a chain of thought will succeed *before a single token is generated*, and beats a BERT classifier reading the generated tokens, which they attribute to the latter relying on shallow linguistic cues. Our mixture result reproduces that ordering and our within-domain result inverts it, which is the disagreement §7.1 is about: for a VLM inside one workload the signal read after generation is the stronger one. Every probe in this literature predicts whether the model will succeed. None predicts whether an intervention would change the outcome, because for text LLMs the intervention is extended reasoning over an unchanged input and the two questions nearly coincide. For a VLM they do not, and §3.3 shows they come apart empirically.

**On the status of these sources.** Adaptive visual acquisition and activation-based routing are both moving quickly, and most of the work closest to ours is recent and not yet peer reviewed, including [11], whose optimality argument §8 transposes. We state its assumptions explicitly and prove our own version rather than inheriting the

| method           | reads           | when           | trained by |
|------------------|-----------------|----------------|------------|
| VisionThink [1]  | special token   | during         | GRPO       |
| AdaptVision [2]  | tool call       | during         | DTPO       |
| AwaRes [3]       | policy output   | during         | SFT+GRPO   |
| GlimpsePrune [4] | glimpse token   | $\frac{2}{3}L$ | a head     |
| entropy baseline | output scores   | after          | n/a        |
| image size       | file header     | before         | n/a        |
| <b>ours</b>      | residual stream | layer $\ell$   | n/a        |

Table 1: What each method reads and what it costs to train. “During” means the decision is a generated token, available only once decoding starts; “before” means the decision needs no forward pass at all, which is the baseline §5.1 finds hardest to beat. The three trained policies also need a supervisor: an LLM judge, GPT-4o, and a LLaMA-3.3-70B judge with an oracle grounding model respectively. GlimpsePrune prunes tokens rather than choosing resolution, and is listed because it is the one method that also reads an intermediate layer.

result, and where we report what a system does we report what its paper reports.

**Where this sits.** Reproducing GRPO training at 7B is not affordable here, so we do not claim a numeric comparison against the systems below; what can be stated precisely is what each one reads, when it decides, and what supervision it needs (Table 1).

The distinction that matters for cost is the third column. Three of the four learn a decision the model *emits*, which requires generation to begin; ours and GlimpsePrune’s read a state the forward pass has already computed. We note that GlimpsePrune measures its own overhead end to end and reports prefill at 69.1% of baseline, so the discipline of charging for the decision is not unique to this work, only unusual among the resolution-choosing methods.

**Calibrated cascade routing.** A separate line takes the routing score as given and asks how the threshold should be set. Kotte [11] argues the threshold should not be tuned at all: calibrate the uncertainty signal to a probability by isotonic regression, and threshold policies on that probability are cost-optimal under three stated assumptions, of which the paper singles out the invariance of the large model’s accuracy to which queries are escalated as the most restrictive and validates it in its own setting. Ruan et al. [12] set gates by an exact lower bound on the survival rate of successful cases, so the policy carries a recall guarantee and cannot degenerate into never escalating, and Tayebati et al. [13] obtain distribution-free coverage for a three-way answer/hedge/abstain split. §8 adopts the first and shows why its central assumption fails for a visual cascade.

**Attacks on the allocation mechanism.** Liu et al. [15] learn a universal trigger, confined to an image border, that

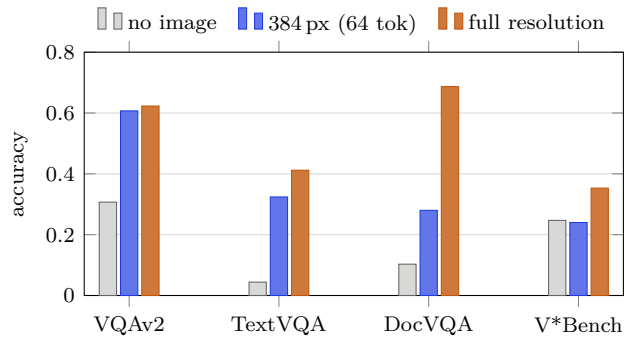


Figure 3: Accuracy per dataset at three information levels, 1000-example pilot. The gap between the blue and orange bars is everything escalation can ever buy, and it ranges from 1.6 points on VQAv2 to 40.7 on DocVQA. On V\*Bench the thumbnail matches the blind baseline: at 384 px its small objects are simply gone.

lowers a weak model’s confidence and so forces deferral to the expensive model: the provider pays and accuracy metrics never register it. Their attack is metric-agnostic because it flattens the output distribution itself, degrading perplexity, token-margin and verbalised confidence alike; Sun et al. [16] report the same failure mode for text cascades. Both target signals read *after* generation, which is where every visual escalation method we have surveyed reads.

**Cost measurement.** Zhan et al. [8] profile on-device VLM energy and find average power to be a model fingerprint, so energy reduces to time; their headline that decode dominates 86 to 97% of it is measured under prompts eliciting hundreds of output tokens, and their own Table 3 shows that fraction collapsing to 0.04 for short answers. Shin et al. [10] decompose sub-3B VLM latency by component on Jetson hardware and report that theoretical reductions do not predict on-device cost. We adopt their decomposition and reach the same conclusion by a different route in §3.2.

## 3 Setup

### 3.1 Configurations and oracle

For each example  $(I, q, a^*)$  we run a set  $\mathcal{C}$  of visual configurations and record, for every  $c \in \mathcal{C}$ , the answer, its correctness, the model’s per-step generation scores, and hardware measurements. Configurations comprise a blind baseline, low-resolution previews at two sizes (ANSWER-LOW), a resolution-capped full pass, one preview+crop per cell of a  $2 \times 2$  grid (CROP), and preview+OCR transcript variants (OCR). The full pilot is 1000 examples across VQAv2, TextVQA, DocVQA and V\*Bench, giving 13 000 instrumented generations.

**Where resolution matters is a property of the benchmark.** Figure 3 is the observation the rest of the paper has to be read against. Cutting the input to 64 visual tokens costs VQAv2 1.6 accuracy points (0.623 to 0.607), and 31% of it is answered correctly with no image at all; DocVQA loses 40.7 points to the same cut, and correctness flips between the two passes on 50% of its queries against 8% for VQAv2. This is the sub-billion transposition of what [1] report at 7B, where a  $4\times$  token reduction is nearly free outside OCR-heavy benchmarks. That a third of VQAv2 is answered with no image at all is also not an artefact of our pilot: Wang et al. [20] construct inputs whose character-level semantics contradict their global visual pattern and find state-of-the-art VLMs consistently resolving the conflict in favour of the text, with visual recognition degrading as semantic complexity rises. A benchmark whose questions are answerable from language priors leaves an escalation policy nothing to allocate, whatever it reads. The consequence for routing is immediate: in any mixed corpus the escalation signal is entangled with dataset identity from the start, and §7 measures how much of the probe’s apparent skill is exactly that.

Correctness uses the metric each benchmark defines (VQA accuracy for VQAv2 and TextVQA, ANLS for DocVQA, exact match for V\*Bench multiple choice) and is recomputed offline from stored answers, so changing the metric never requires re-running the model.

The oracle labels each example with the cheapest configuration that answered correctly, under

$$J(c) = w_e \mathbf{1}[\hat{a}_c \neq a^*] + \lambda_t t_c + \lambda_e E_c + \lambda_m M_c + \lambda_v V_c, \quad (1)$$

with  $t$  latency,  $E$  energy above a measured idle baseline,  $M$  peak memory and  $V$  visual tokens. Weights in Eq. 1 are calibrated so each resource term contributes 0.03 to 0.09 against an error weight of 1.0: the oracle requires correctness first, then discriminates on measured cost.

**What the weights can and cannot change.** Calibrating them so each resource term lands in a narrow band is a convention, not an argument, and Proposition 1 minimises this function, so a reader is entitled to ask what a different choice would do. Two answers, one structural and one measured.

Structurally, **oracle accuracy cannot depend on the weights at all.** The oracle ranks only among configurations that answered correctly, so no setting of  $\lambda$  can make a wrong answer preferable to a right one, and the set of solvable examples is fixed before any weight is read. Sweeping every weight over four orders of magnitude confirms it exactly: accuracy stays at 0.694 to within  $10^{-9}$ .

For the same reason  $w_e$  **is inert in the labelling:** it is added to options the oracle has already discarded, so 100% of labels survive a  $10\,000\times$  change in it. The error weight matters only through the ratio  $V = w_e/\lambda_t$  in the serving rule of §8, where it is swept rather than

fixed. Our own calibration story, phrased as resource terms measured “against an error weight of 1.0”, therefore describes something the labelling never consults.

What the weights do choose is *which* correct action is cheapest. A hundredfold rise in  $\lambda_t$  moves the oracle’s mix from 16% to 34% CROP and from 11% to 2% OCR, leaving 75% of labels unchanged. That is not a fragility, it is the budget-dependence this paper reports as a finding: when visual tokens are dear, a preview plus a transcript beats a high-resolution crop. The reader should treat every accuracy claim here as weight-free and every action-mix claim as conditional on a stated budget.

Two further properties of Eq. 1 are worth stating. Being linear in the resources, it expresses only finitely many policies: sweeping  $\lambda_v$  over its whole range on our pilot yields exactly two, CROP below a crossover and ANSWER-LOW above. And a policy conditioned on a probe must be charged for the probe, so an escalated query pays both passes throughout this paper.

### 3.2 What a pass actually costs

Every policy comparison in this paper is a cost comparison, so the cost model has to be measured rather than assumed. We time each stage of one pass with explicit CUDA synchronisation, following the component decomposition of [10]. Two standard proxies turn out to mispredict, and both matter for what follows.

**The vision encoder is half the visual cost.** Table 2 attributes latency on three models spanning  $8.6\times$  in parameters and two distinct vision encoders.

| model         | encoder | extra prefill | encoder share |
|---------------|---------|---------------|---------------|
| SmolVLM-256M  | 52.5    | 55.0          | 49%           |
| SmolVLM-500M  | 52.4    | 53.5          | 50%           |
| SmolVLM2-2.2B | 193.6   | 252.8         | 43%           |

Table 2: Visual cost split at 768 px, in milliseconds.

This bounds a whole family of methods. Post-encoder pruning [4] can only recover the prefill half, because the encoder has already run when tokens are scored; reducing input resolution recovers both. **Token pruning is capped at roughly half the available saving**, and the ratio holds across the parameter range. It also tells us what a probe reading mid-prefill can and cannot save, which §5 makes precise.

**Decode latency follows depth, not size.** SmolVLM2-2.2B emits the same three-token answer in 47.2 ms against 77.0 ms for the  $4.4\times$  smaller 500M model, because it has 24 layers rather than 32; per-layer cost is roughly constant at 2.0 to 2.4 ms. Choosing an edge model by parameter count optimises the wrong variable, and a cost model linear in parameters, as the power fit of [8] is, will not transfer here.

**Cache reuse cannot refund the cheap pass.** Our accounting charges an escalated query for both passes with no reuse, and the obvious objection is that a serving stack would not. VLCache [17] reports 1.2 to 16 $\times$  time-to-first-token speedups by reusing encoder and KV caches across multimodal requests. Two facts close this, and neither needs an assumption about what a stack achieves.

First, escalation is a cache miss by construction: VL-Cache keys its cache on a hash over the input pixels, and escalating submits different pixels. The same holds for the text-side systems this line grew from: CachedAttention’s hierarchical store [21] exists to reuse the key-value cache of *historical tokens* across the turns of one conversation. That literature refunds repetition, and escalation is the opposite of repetition.

Second, and independently, decode is uncacheable. Reading output entropy requires the cheap pass to *generate*; generation produces the answer, differs per query, and no cache returns it. Writing  $c$  for the fraction of encoder and prefill a cache refunds, the probe’s saving per escalated query is  $(1 - c)(1 - \ell/L)t_{\text{prefill}} + t_{\text{decode}}$ , which runs from 103.1 ms at  $c = 0$  to a floor of 64.7 ms at a perfect cache, still 63% of the uncached saving. No refund fraction erases it.

**Energy is not reported.** Our NVML integration disagrees by 18 to 96% between configurations spending identical visual tokens, while their latencies agree within 4 to 5%; a standing check refuses to report joules until that is repaired. This also bears on reading [8], whose headline, decode dominates 86 to 97% of inference energy, is measured under prompts eliciting hundreds of output tokens. Their own Table 3 shows the decode fraction collapsing to 0.04 under short-answer prompts, the regime every dataset here occupies. All costs in this paper are therefore latencies.

### 3.3 The three routing questions

Let  $y_{\text{cheap}}$  and  $y_{\text{full}}$  be the correctness of the low-resolution and full-resolution passes. Three distinct binary targets can be formed over the same examples:

$$Q_1 : y_{\text{cheap}} \quad (\text{is the cheap pass correct?}) \quad (2)$$

$$Q_2 : \neg y_{\text{cheap}} \wedge y_{\text{full}} \quad (\text{does escalation recover?}) \quad (3)$$

$$Q_3 : \bigvee_{c \in \mathcal{C}_{\text{route}}} y_c \quad (\text{answerable at all?}) \quad (4)$$

The escalation literature conditions on  $Q_1$ . The quantity that decides whether compute is well spent is  $Q_2$ , evaluated on the subset where the cheap pass failed.  $Q_3$  separates queries no visual operation can rescue.

## 4 Confidence answers the wrong question

Mean token entropy over the generated answer predicts  $Q_1$  well and  $Q_2$  poorly. Table 3 reports AUROC on held-out

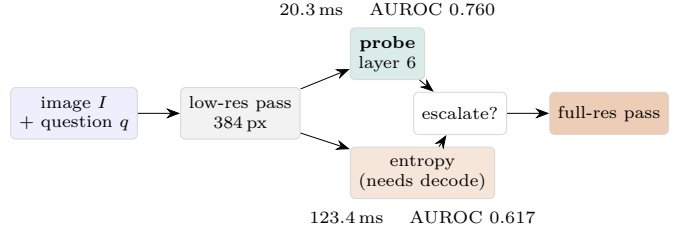


Figure 4: Two allocation signals read from the same low-resolution pass. Output entropy requires the pass to complete and generate; the probe reads the residual stream at layer 6 of 32 and needs no decode.

| target                 | output entropy | pre-generation probe |
|------------------------|----------------|----------------------|
| $Q_1$ correct          | 0.758          | 0.822                |
| $Q_2$ escalation helps | 0.617          | <b>0.760</b>         |
| $Q_3$ answerable       | 0.539          | 0.681                |

Table 3: AUROC by target. Entropy is scored with the sign that favours it; read with the  $Q_1$  sign it scores 0.372 on  $Q_2$ , because the sign flips between the two questions.

folds,  $n = 1000$  overall and  $n = 617$  for  $Q_2$ .

The sign flip is the mechanism. A *confident* wrong answer is a knowledge failure that more pixels do not fix; an *uncertain* wrong answer is often a perception failure that they do. Splitting the 617 failures at their median entropy, escalation recovers 25% of confident failures and 42% of uncertain ones.

The two effects reinforce for a policy. Writing the net benefit of escalating as

$$G = \mathbf{1}[\neg y_{\text{cheap}} \wedge y_{\text{full}}] - \mathbf{1}[y_{\text{cheap}} \wedge \neg y_{\text{full}}], \quad (5)$$

$\mathbb{E}[G]$  rises monotonically across entropy quintiles, from +5% to +41.5% (Fig. 5). Escalation is worth eight times more in the top quintile than the bottom.

Escalation is also not monotone: 4.1% of queries are answered correctly at 384 px and wrongly at full resolution. No surveyed method models this risk.

## 5 A pre-generation probe

Following [5, 6], we read the residual stream  $h^{(\ell)} \in \mathbb{R}^d$  at the last prompt position (after the instruction, before any token is generated) and fit the difference-of-means direction

$$w^{(\ell)} = \mu_+^{(\ell)} - \mu_-^{(\ell)}, \quad s(h) = \frac{(h - \bar{\mu}^{(\ell)})^\top w^{(\ell)}}{\|w^{(\ell)}\|}, \quad (6)$$

where  $\mu_{\pm}$  are class centroids and  $\bar{\mu}$  their midpoint. Eq. 6 is deliberately linear: the question is whether the signal is linearly accessible, not how well a classifier can be tuned. Stated that way it is a framing rather than a defence, so we measured what capacity buys (Table 4).

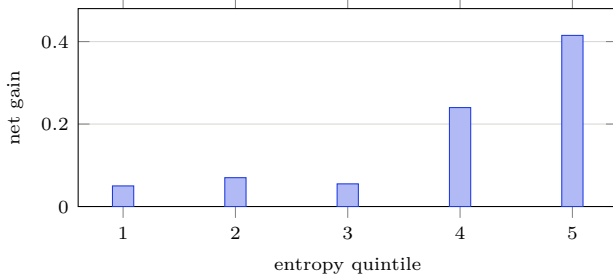


Figure 5: Net benefit of escalating, by entropy quintile ( $n = 1000$ , 200 per bin). AUROC of entropy for predicting net benefit is 0.734.

| probe family            | pooled mixture | within DocVQA |
|-------------------------|----------------|---------------|
| difference of means     | <b>0.749</b>   | 0.576         |
| logistic                | 0.679          | 0.570         |
| random Fourier features | 0.699          | 0.596         |
| shrunk centroid (LDA)   | 0.701          | 0.579         |
| output entropy          | 0.735          | <b>0.656</b>  |

Table 4: AUROC on the joint escalation target, 20 resampled folds, layer 6. Random Fourier features give a smooth non-linear boundary without a bespoke architecture; the shrunk centroid is the covariance correction the plain difference of means omits. The within-DocVQA column is fitted on the 1200-page pilot at its full training size; the learning curve of Fig. 12 reports 0.572 for the same probe at 900 training examples over 30 resamples, and Table 14’s 0.519 is the different quantity of fitting inside each of the four mixture datasets on 210 examples apiece.

On the mixture the two-centroid difference *beats* every fitted alternative by 0.048 or more, so the paper’s probe is not an undertrained stand-in for something better; fitting parameters on 700 examples in 960 dimensions costs more than it recovers. And within a domain no family closes the gap to output entropy: the best of the four reaches 0.596 against entropy’s 0.656 [0.645, 0.666]. **The within-domain negative is a property of the representation, not of linear models**, which is the objection this table exists to answer. It remains one family of non-linearities rather than a proof, but it is the family a reviewer would reach for first. Fitting is a single pass over cached activations, and the resulting score is the one consumed by the decision node of Fig. 4.

**Depth.** Fig. 6 sweeps  $\ell$ . The depth used everywhere in this paper is selected on the validation fold, which independently picks layer 6 (validation AUROC 0.774 on the conditional target, 0.790 on the joint one); the test curve below is illustration, not selection. The signal is absent at  $\ell = 0$  by construction, the last prompt token is textual and has attended to nothing, and reaches its ceiling by  $\ell = 3$ . Bootstrapped, layer 6 gives 0.760 [0.668, 0.849] and layer 23 gives 0.760 [0.652, 0.853]: indistinguishable.

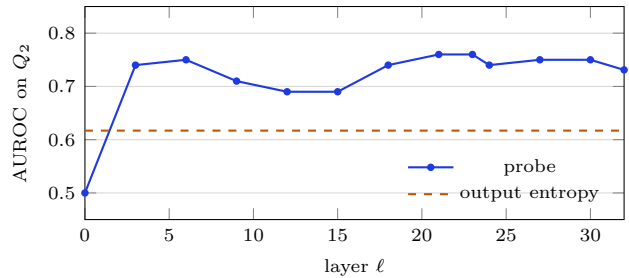


Figure 6: Probe AUROC on  $Q_2$  against depth, single fit per depth on the reported fold; the 0.760 quoted in the text is the bootstrapped estimate at layer 6 over the same fold, which the coarse sweep here renders as 0.750. The ceiling is reached by layer 3 and never improves, so the escalation decision can be made after one sixth of the language model.

**Cost.** Decomposing one pass with CUDA synchronisation around each component, a 384 px pass costs 11.0 ms in the vision encoder, 48.1 ms in prefill and 67.8 ms in decode. Reading at layer  $\ell$  needs the encoder plus a fraction of prefill,

$$t_{\text{probe}}(\ell) = t_{\text{enc}} + t_{\text{proj}} + \frac{\ell}{L} t_{\text{prefill}}, \quad (7)$$

giving 20.3 ms at  $\ell = 6$  against 123.4 ms for a complete pass.

That ratio is *not* the saving, and it is worth being explicit about why. Entropy requires the cheap pass to finish, but so does answering, whenever the policy decides *not* to escalate. The probe therefore saves only on escalated queries, where the remainder of the cheap pass is abandoned. With escalation rate  $\rho$  and full-resolution cost  $t_{\text{full}}$ ,

$$C_{\text{entropy}} = t_{\text{cheap}} + \rho t_{\text{full}}, \quad (8)$$

$$C_{\text{probe}} = (1 - \rho) t_{\text{cheap}} + \rho (t_{\text{probe}} + t_{\text{full}}). \quad (9)$$

At our observed  $\rho \approx 0.30$ , Eq. 9 gives 185.3 ms against 154.3 ms: a 17% saving, rising to 31% if every query escalates and falling to 7% at  $\rho = 0.10$ . The probe is cheaper, but by a sixth, not by the 6 $\times$  that comparing  $t_{\text{probe}}$  from Eq. 7 against a full pass would suggest. We state this explicitly because the read-cost ratio is the number an efficiency paper is tempted to report.

## 5.1 A better ranking is not a better policy

AUROC says a signal orders queries well; it does not say a policy built on it serves better answers per unit of compute. Building both policies at matched escalation rates and charging them by Eq. 9 exposes a trap.

A probe trained on  $Q_2$  (the *conditional* target, defined only on queries the cheap pass failed) produces a policy that is 8.5 accuracy points *worse* than the entropy threshold it beat on AUROC ( $-0.085$  [ $-0.130$ ,  $-0.040$ ], paired). The reason is that a deployed policy must decide on every

| policy (escalating 30%)        | accuracy     | latency      | AUROC on $G$ |
|--------------------------------|--------------|--------------|--------------|
| always cheap                   | 0.385        | 123.4        | n/a          |
| entropy threshold              | 0.480        | 182.2        | 0.751        |
| probe on $Q_2$                 | 0.395        | 151.7        | 0.259        |
| <b>probe on <math>G</math></b> | <b>0.510</b> | <b>154.3</b> | <b>0.761</b> |
| always full                    | 0.530        | 206.0        | n/a          |
| oracle                         | 0.580        | 139.6        | n/a          |

Table 5: End-to-end policy comparison on the held-out fold, latency in milliseconds. Against the entropy threshold, the joint-target probe differs in accuracy by  $+0.030$   $[-0.005, 0.070]$ , indistinguishable, and in latency by  $-27.8$   $[-40.7, -15.0]$  ms, a significant 15% saving.

| policy                          | escalates | accuracy     | latency      |
|---------------------------------|-----------|--------------|--------------|
| always cheap                    | 0%        | 0.385        | 123.4        |
| self-report, “ANSWER or ZOOM”   | 7%        | 0.385        | 137.9        |
| self-report, “detailed enough?” | 22%       | 0.405        | 169.8        |
| entropy threshold               | 20%       | 0.465        | 164.7        |
| <b>probe</b>                    | 20%       | <b>0.475</b> | <b>144.0</b> |
| always full                     | 100%      | 0.530        | 206.0        |

Table 6: Untrained self-report against uncertainty-based routing, 200 held-out examples, latency in milliseconds. Against the stronger self-report the probe is  $+0.070$   $[0.020, 0.120]$  more accurate and 25.8  $[11.9, 39.2]$  ms cheaper, both paired and both significant.

query, not only on failures, and the quantity it needs is the joint event  $G$ : escalation helps only when the cheap pass would fail *and* full resolution would succeed. Entropy predicts  $G$  at 0.751 because both factors rise with it. The  $Q_2$ -trained probe predicts  $G$  at 0.259, far below chance, because it never saw a successful cheap pass.

Retraining the same probe on  $G$  fixes it (Table 5).

And the  $Q_2$  row is the more useful result: a probe can beat a baseline on a conditional target and lose decisively as a policy. Papers that report ranking quality without building the policy risk exactly this.

**Asking the model directly is worse than reading its uncertainty.** The escalation mechanism published systems train is a self-report: shown a downsampled image, the model states whether it needs the full-resolution one. We cannot reproduce VisionThink’s GRPO at sub-1B, that is the premise of this work, but we can implement its *mechanism* without the training, and ask what the untrained version is worth. Two phrasings were tried, both eliciting a one-word judgement (Table 6).

The first phrasing escalates 7% of queries and recovers nothing: its accuracy is exactly that of never escalating, so the compute is pure waste. The second escalates 22% and buys two accuracy points, where routing on entropy at the same rate buys eight and routing on the probe buys nine at lower cost. The self-report is not merely weaker; it escalates as often as the alternatives and selects the wrong queries.

This bounds what the untrained mechanism delivers and, read the other way, quantifies what the training in [1, 2, 3] is buying. It is not a comparison against those systems, which would require reproducing their training at a scale where it is not affordable.

**One prediction that can be checked against a 7B system without running one.** We have declined a numeric comparison with the trained policies throughout, because reproducing GRPO at 7B is the thing this setting cannot afford. That refusal leaves our structural findings unconnected to the systems we position against, and one of them can be connected without any training at all.

The finding is that an escalation policy degenerates towards always escalating exactly where the cheap pass fails on most queries, because the quantity it thresholds saturates while the recoverable share does not. We measure it twice: the correctness-calibrated rule sends every query to full resolution on our 256M model against 54% on the 500M (§8), and on ChartQA the cheap pass is wrong on 80% of queries (Table 12).

VisionThink [1], trained with GRPO on Qwen2.5-VL-7B, reports the same behaviour on the same benchmark and says so plainly: on ChartQA their model consumes more time than their QwenRL baseline, and 101.4% of its visual tokens, because it “identifies that most questions cannot be answered correctly at low resolution and thus autonomously requests high-resolution images”, so its total visual-token count exceeds the baseline’s. They report the opposite on DocVQA, where they are more than twice as fast.

**Training the decision does not remove the degeneracy; it relocates it.** The published system escalates almost everything on the corpus where the cheap pass mostly fails, which is what an uncalibrated threshold does and what our gain-calibrated rule is constructed to avoid. We cannot say whether our correction would help theirs, since their policy is a generated token and not a threshold on a score. What we can say is that the failure mode we characterise at sub-billion scale is visible in a 7B system’s own reported numbers, which is the evidence a scale-limited paper can offer about scale.

**The whole frontier, not one operating point.** A single escalation rate flatters or damns arbitrarily. Sweeping it (Fig. 7) gives a cleaner statement: on this fold every entropy operating point is dominated. Of the fourteen policies evaluated, six lie on the accuracy-cost Pareto front, four of them probe-based and none entropy-based.

At matched accuracy the saving is 12% at 0.46 and 22% at 0.52. The 0.52 point matters most: the probe reaches 0.525 at 161.5 ms where always running at full resolution reaches 0.530 at 206.0 ms, within half a point of the accuracy ceiling for 78% of its cost, while the entropy policy at the same accuracy costs 206.9 ms, marginally *more* than simply always escalating.

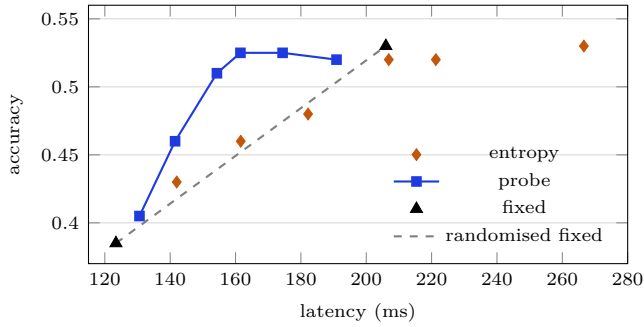


Figure 7: Accuracy against latency as the escalation rate sweeps from 10% to 70%, on the reported fold. The dashed chord is what randomising between the two fixed policies reaches with no signal. On this fold two entropy points sit marginally above it; resampled, the best entropy operating point is indistinguishable from the chord and every other sits below it (Table 7). Four probe points lie on the Pareto front, together with the two fixed policies at its ends. Fig. 8 tests how much of this survives resplitting, and Table 8 adds the two comparators that supersede this comparison: a free image descriptor and a policy that aborts at random.

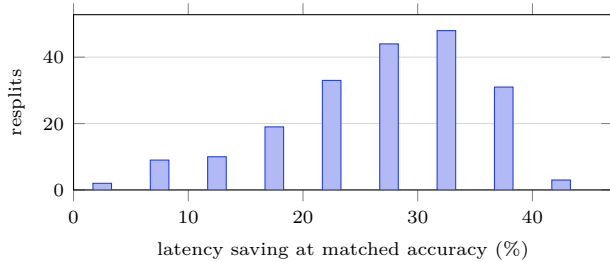


Figure 8: The probe’s latency saving over entropy at matched accuracy, across 200 stratified resplits. The distribution is entirely positive; the single-fold figure of 12 to 22% quoted above is a conservative corner of it.

**How much of that is the fold?** Universal dominance is a statement about six points on 200 test examples, which is not a scale at which a categorical claim should be made. We therefore refit everything downstream of the split (probe direction, thresholds, calibration) on 200 stratified resplits.

The universal claim does not survive: all seven entropy operating points are dominated in only 52% of splits. What survives is stronger than that number suggests. On average 0.83 of seven entropy points escape domination, and when one does it wins by a median of 0.010 accuracy, two queries out of two hundred, with 85% of survivors winning by 0.02 or less. Meanwhile the probe holds 90% of the Pareto front, and at matched accuracy it is cheaper in 200 of 200 splits, by a median of 27.8% (Fig. 8).

The honest summary is that the direction of the effect is robust and its universality is not. We report the distribution rather than the fold because the fold is what a

| gap to the hull  | $V=400$       | 800    | 1600          | 3200   |
|------------------|---------------|--------|---------------|--------|
| probe, mixture   | <b>+0.054</b> | +0.051 | +0.033        | +0.008 |
| entropy, mixture | −0.001        | −0.016 | <b>−0.032</b> | −0.026 |
| ladder, DocVQA   | +0.042        | +0.029 | +0.017        | +0.006 |

Table 7: Accuracy above what randomising between fixed configurations reaches at the same expected latency, 30 resampled folds; every probe and ladder interval excludes zero and every entropy interval excludes any gain above +0.005. Prices here are per configuration; Table 8 repeats the mixture rows under the per-example prices of \$7, where the probe’s margin narrows from +0.053 to +0.033 at  $V=400$  and the ranking of the policies is unchanged.

single split of 1000 examples can support.

**The baseline that needs no signal.** Both comparisons above are between adaptive policies, and that is not the bar a deployment sets. Any point on the segment between two fixed policies is reachable by randomisation: serve a fraction of queries at the dear configuration and the rest at the cheap one. The convex hull of the fixed policies is therefore the floor an adaptive policy must clear, and no comparison against a single endpoint establishes it. Table 7 reports each policy’s accuracy minus the hull accuracy at its own expected latency, paired over 30 resampled folds.

## 6 Which signal pays depends on the traffic

This section prices each candidate signal against the randomisation hull, first on the benchmark mixture and then inside single workloads. The two answers disagree, and the disagreement is the paper’s central claim.

The comparison is not an invention of ours: random allocation is a standard baseline in text routing, where [5] plot their cascades against it, and it is absent from the visual escalation evaluations we survey. Nor does the verdict depend on how the operating point is chosen. Sweeping quantile rates instead of preferences, entropy’s best point is indistinguishable from the chord (+0.002 [−0.002, +0.005] at a 10% rate) and every other rate sits significantly below it, while the probe clears the chord at all seven rates (+0.006 to +0.056).

Three readings, each load-bearing. The entropy threshold, the baseline this literature measures against, does not clear the hull here at any operating point of either parameterisation: reading it requires completing the pass it prices, and the read cost eats the routing gain. The probe clears it at every tight and moderate budget, by most where the budget binds, and only at the loosest preference under per-example prices does its interval span zero (−0.001 [−0.004, +0.002] at  $V=3200$ ). Table 8 locates that margin: not in the mid-prefill abort, since randomising with the same abort clears nothing, but in

| gap to the hull                 | V=400         | 800    | 1600   | 3200   |
|---------------------------------|---------------|--------|--------|--------|
| <i>per-configuration prices</i> |               |        |        |        |
| probe                           | +0.053        | +0.050 | +0.033 | +0.001 |
| image size (free)               | <b>+0.061</b> | +0.050 | +0.024 | +0.009 |
| entropy                         | -0.001        | -0.012 | -0.028 | -0.022 |
| random + abort                  | -0.0005       | -0.007 | -0.001 | -0.001 |
| <i>per-example prices</i>       |               |        |        |        |
| probe                           | +0.033        | +0.032 | +0.021 | -0.001 |
| image size (free)               | <b>+0.034</b> | +0.027 | +0.011 | +0.002 |
| entropy                         | -0.001        | -0.002 | -0.014 | -0.017 |
| random + abort                  | -0.0001       | -0.006 | -0.001 | -0.001 |

Table 8: Accuracy above the randomisation hull at matched expected latency, 30 resampled folds, under both costings. Every image-size interval excludes zero; no random-plus-abort interval does. The probe rows are recomputed on resamples independent of those behind Table 7 and agree with it to within 0.007.

the signal, which §7 shows is largely a domain effect and which a free image descriptor supplies as well. And on the homogeneous pilot the ladder clears the hull where binary escalation does not (§10): the graded action space, not the signal, beats the coin flip there. Adaptive escalation pays where its read is cheap or its actions are graded, and not otherwise.

**Two comparators that are cheaper still.** A hull comparison answers whether a signal beats no signal. It does not answer which part of an adaptive policy is doing the work, and two candidates cost less than the probe.

The first is the free one. §7 finds raw image size scoring 0.748 against the probe’s 0.761, and a signal read from the file header does not merely rank almost as well: it prices differently. A policy reading it decides before anything runs, so an escalated query never pays for a cheap pass it discards, and its cost structure is exactly that of the randomisation it is being compared against. The second isolates the mechanism rather than the signal: escalate on a coin flip, but abandon the cheap pass mid-prefill as the probe does. If the probe’s margin is owed to the abort, this should reproduce it (Table 8).

Both answers are inconvenient for the apparatus this paper built. **On this serving model a signal costing nothing to read clears the hull at every operating point of either costing**, and every one of those clearances survives family-wise correction. Paired against the probe over the same resamples, the free signal differs by at most 0.010 either way, and of the preferences where one of them nominally wins, none survives correction. That equivalence is specific to this model: Table 9 repeats the comparison on a 256M model, where the probe keeps an advantage at tight budgets and the free descriptor does not clear under per-example prices at all. On this mixture the pre-generation probe buys nothing that the image dimensions do not already supply, and the cases where it

| gap to the hull, SmolVLM-256M   | V=400         | 800    | 1600   | 3200   |
|---------------------------------|---------------|--------|--------|--------|
| <i>per-configuration prices</i> |               |        |        |        |
| probe                           | <b>+0.046</b> | +0.013 | -0.004 | -0.003 |
| image size (free)               | +0.033        | +0.007 | +0.004 | +0.002 |
| entropy                         | -0.000        | -0.056 | -0.018 | -0.008 |
| random + abort                  | -0.004        | -0.004 | -0.001 | -0.001 |
| <i>per-example prices</i>       |               |        |        |        |
| probe                           | <b>+0.019</b> | +0.005 | -0.004 | -0.003 |
| image size (free)               | -0.001        | -0.002 | -0.002 | -0.001 |
| entropy                         | -0.000        | -0.046 | -0.018 | -0.008 |
| random + abort                  | -0.003        | -0.003 | -0.001 | -0.001 |

Table 9: The mixture hull comparison repeated with a 256M serving model, 30 resampled folds. Two results replicate and one does not.

appears to are the cases the correction removes.

**And the abort mechanism, alone, buys nothing at all.** Random escalation with the same mid-prefill abandonment sits at or below the hull everywhere, from -0.0005 to -0.007 swept by preference and to -0.022 swept by rate. The margin the probe opens is therefore bought by its signal and not by its read cost, which corrects an attribution an earlier version of this section made in the opposite direction. What the signal knows, given §7, is which dataset the query came from, and image size knows the same thing for free.

**The same comparison on a second sub-billion model, and where it stops replicating.** Every number above is measured with one 500M serving model, which is thin evidence for a claim about a scale regime. We repeated the whole comparison on SmolVLM-256M over the same 1000 images, the same activations protocol and the same two costings (Table 9).

Two of the three findings replicate and the third does not. Output entropy still never clears the hull, at any preference of either costing, and randomising with the probe’s mid-prefill abort still clears nothing, so the margin still belongs to the signal rather than to the read mechanism. What does not replicate is the equivalence between the probe and the free descriptor. Under per-configuration prices the free descriptor clears at all four preferences and the probe at two; under the per-example prices we argue are the honest ones, the free descriptor clears at none and the probe still clears at the two tightest budgets, beating it by 0.019 [0.014, 0.024] at V=400.

**The claim that a file-header read matches a probe on activations is therefore specific to the 500M model, not to the scale.** On the smaller model the pre-generation read earns its cost where the budget binds, which is the regime an efficiency paper cares about most.

The workload half of the inversion was also measured on one model, so we repeated it too: on DocVQA with the 256M model the free descriptor clears nothing at either rule or any preference, as on the 500M, while the entropy

ladder clears at one preference of four rather than three. The ordering survives the change of model and the margin does not. Across the four workload-model pairs we then had, the free descriptor cleared nothing in any. A fifth pair, added later with a third lineage, is the exception and is analysed in §10: there it clears while remaining at chance on the target, which separates ranking from cost allocation.

We state the general claim as the weaker one both models support, that a free descriptor is a comparator this literature has to beat and that entropy does not beat it either, and we do not claim the equivalence beyond the model on which it was measured.

### 6.1 Is the free descriptor a signal or a detector?

A free descriptor that clears the hull across benchmarks and fails inside every one of them admits two readings, and they differ for anyone deciding whether to deploy it. Either image size carries information about whether more pixels would help, distributed unevenly across benchmarks, or it is a benchmark detector that works only because benchmarks differ in escalation value. Separating them needs a stratum where size still varies and provenance is mixed, and the pilot contains one: images between 1024 and 1936 px,  $n = 288$ , standard deviation 230 px, 15% minority provenance.

Inside it the descriptor predicts the source dataset at AUROC 0.976 and the escalation gain at 0.579. The value gap it would need to be reading survives in full: escalation repairs 36% of the DocVQA pages in the stratum against 13% of the TextVQA ones. **Image size is a provenance detector.** Its pooled 0.730 on the escalation gain is bought by separating benchmarks that differ in value, not by ranking queries within one, which is the same finding §7 reports for the probe and the reason both fail inside a workload.

The consequence is a deployment condition rather than a caveat. A router whose traffic couples image size to provenance, as a mixture of public benchmarks does, can read the file header and skip the model. A router whose traffic varies in content at similar sizes cannot, and nothing in this paper tells it what to read instead beyond the model’s own uncertainty.

**And an upper bound on what the confound can be worth.** If the signals above are reading dataset identity, the natural ceiling is a policy handed that identity outright. We give one the label and let it pick a rate per dataset, escalating a uniformly random subset inside each. Selection within a dataset stays random by construction, so the policy knows *which benchmark* and nothing whatever about the query (Table 10).

The bound dominates every signal in this paper at every preference, while using no information about any individual query. What it does with the label is blunt,

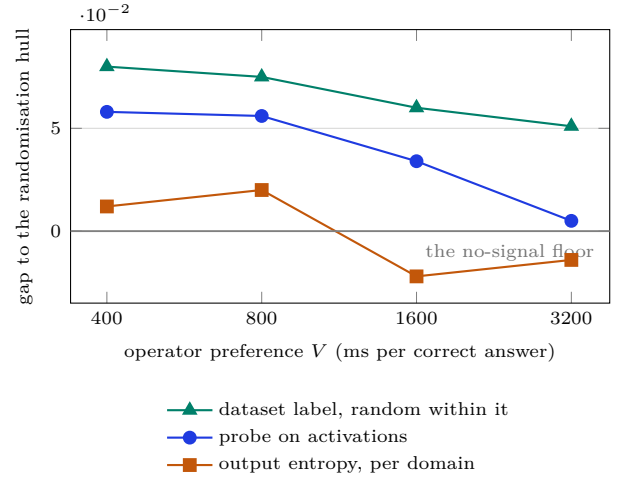


Figure 9: The ceiling, on the same quantity as the floor. Each curve is a policy’s gap to the randomisation hull on the four-dataset mixture, swept over the operator preference. A policy handed *only* which dataset a query came from, and escalating at random inside it, sits above every signal we read from the model or the file at every preference. It knows nothing about any individual query. **On mixed traffic the adaptive gain available is therefore close to the gain from knowing which benchmark is being served**, and a per-query signal is competing for what remains. Output entropy falls below the floor at the two loosest preferences, which is the mixture result §6 reports and §7.1 inverts.

and that is the point: unable to choose *which* queries, it escalates almost the whole of the domain where escalation pays and almost none of the domain where it does not, 100% of DocVQA against 13% of VQAv2 averaged over preferences, with V\*Bench and TextVQA between them. **On a benchmark mixture, essentially all of the adaptive gain available is the gain from knowing which benchmark you are serving**, and a router that already knows its traffic composition has it for free. That is the sharpest statement of the confound: not merely that the probe correlates with domain, but that domain is the whole of what any of these signals is worth here.

The last row is the complement. Applying one unchanged preference inside each dataset, on output entropy, clears the hull at tight budgets and falls below it at loose ones. Within a domain, once identity is removed, what remains for a per-query signal to earn is small and inconsistently signed, which is §7.1 restated in policy terms rather than in AUROC.

### 6.2 The second regime: inside a single workload

**The free descriptor does not survive leaving the mixture.** If image size routed inside a single workload too, nothing in this paper would justify reading the model

| gap to the hull          | V=400         | 800           | 1600          | 3200          |
|--------------------------|---------------|---------------|---------------|---------------|
| domain label, tuned rate | <b>+0.080</b> | <b>+0.075</b> | <b>+0.060</b> | <b>+0.051</b> |
| probe                    | +0.058        | +0.056        | +0.034        | +0.005        |
| per-domain entropy rule  | +0.012        | +0.020        | -0.022        | -0.014        |

Table 10: What dataset identity alone buys, 30 resampled folds. The first row escalates a random subset within each dataset at a rate tuned per dataset on the test fold, so it is an optimistic bound rather than a deployable policy. The probe row is recomputed on these resamples and agrees with Table 8 to within 0.005.

| gap to the hull, DocVQA | V=400         | 800           | 1600          | 3200   |
|-------------------------|---------------|---------------|---------------|--------|
| ladder, entropy         | <b>+0.041</b> | <b>+0.025</b> | <b>+0.010</b> | +0.001 |
| ladder, image size      | +0.000        | -0.001        | -0.004        | -0.004 |
| binary, entropy         | -0.003        | -0.081        | -0.017        | -0.015 |
| binary, image size      | -0.026        | -0.052        | -0.012        | -0.010 |

Table 11: The same comparison as Table 8, run inside one workload instead of across four. The hull is built from all four fixed rungs, 30 resampled folds, per-example prices. The entropy ladder row is recomputed on independent resamples and agrees with Table 7 to within 0.007.

at all. It does not. On the same 1200 pages, the same verified rungs and the same per-example prices, image size predicts the top-rung gain at 0.508, which is chance, against output entropy’s 0.667; and built into a policy it never clears the hull, sitting at or below it at every preference (Table 11).

**The two regimes require opposite answers, and this is the paper’s central claim.** Across a mixture of benchmarks, escalation value is carried by which benchmark a query came from, a free image descriptor recovers it, and neither output entropy nor a probe on activations earns its read cost. Inside a single workload that between-domain axis is gone: the free descriptor falls to 0.508 to 0.586 AUROC on the escalation target, and what ranks queries by value is a signal read from the model, spent on a graded action space rather than on a binary one. Neither regime is the general case, and a method evaluated only in the first will recommend the wrong thing for the second.

The two halves are measured on eight corpus-model pairs and they do not carry equal weight. *As a ranking claim the inversion is unqualified:* the descriptor is the weaker value ranker on all eight, scoring 0.508 to 0.586 on the escalation target against a model-read signal’s 0.609 to 0.791. *As a policy claim it is not,* and the difference is what §9 is about. The model-read ladder clears the hull on five of the eight by +0.023 to +0.075, against a signal-free slack of at most +0.008 on those pairs; it fails on InfographicVQA read by the 500M model, where the serving model is the binding constraint, and its clearance on InfographicVQA read by Qwen2-VL-2B is withdrawn because a signal-free policy does better there. The descriptor’s own policy clears on the three pairs whose serving model prices resolution

| gap to the hull, ChartQA | V=400  | 800           | 1600   | 3200   |
|--------------------------|--------|---------------|--------|--------|
| output entropy           | +0.020 | <b>+0.032</b> | +0.019 | +0.012 |
| image size (free)        | +0.005 | -0.002        | -0.003 | -0.003 |

Table 12: The surviving rung on 500 ChartQA questions, 30 resampled folds, per-example prices. Every entropy interval excludes zero. The ladder and the binary rule coincide here to within 0.002, because Algorithm 4 left only one rung for either to choose.

steeply, about half of which is budget allocation that needs no signal at all. What we claim generally is the inversion of the *ordering* on the ranking target. Whether that ordering has cash value depends on the price structure of the serving model, and on one pair it reverses outright.

That is also the reading under which our own negative results stop being awkward. The probe fails inside a domain (§7.1) and is redundant across one; output entropy fails across a mixture (Table 8) and is the only signal that works inside one. Both failures are the same fact seen from two sides: a signal is worth exactly the variance it explains that the free features do not.

### A third workload, and the guard that fires on it.

The positive half of the inversion rested on one corpus, so we collected a third: 500 ChartQA questions, scored with the relaxed accuracy the benchmark defines (a numeric answer within 5% of the gold value, exact match otherwise), because grading chart answers by edit distance would award partial credit for being close to a different number.

Step 3 of Algorithm 4 fires before any policy is priced. ChartQA charts have a median longest edge of 800px, so the 1152px and full-resolution targets are upsampling: token counts rise on 2% and 0% of pages respectively, against 99% and 95% on DocVQA. The procedure discards both as duplicates and leaves a two-rung ladder, which is the first time one of this paper’s own guards has rejected a configuration on a corpus it was not written against. What remains is a binary choice, and the comparison is run on that (Table 12).

**The inversion holds as a statement about ranking, and the policy side is decided by price.** The free descriptor is a weak ranker inside every workload we have measured, scoring 0.508 to 0.586 on the escalation gain against a model-read signal’s 0.609 to 0.791 over eight pairs. Its *policy* is another matter and depends on the serving model rather than the corpus, which Table 13 shows and §9 explains. A signal read from the model clears the hull on two of the three, at four preferences of four on ChartQA and three of four on DocVQA, all surviving correction. On InfographicVQA it clears at one preference nominally and that clearance is lost to the correction ( $p = 0.004$ , adjusted 0.108), while at the loosest preference the ladder sits significantly *below* the hull. We read that as the corpus rather than the signal: the model

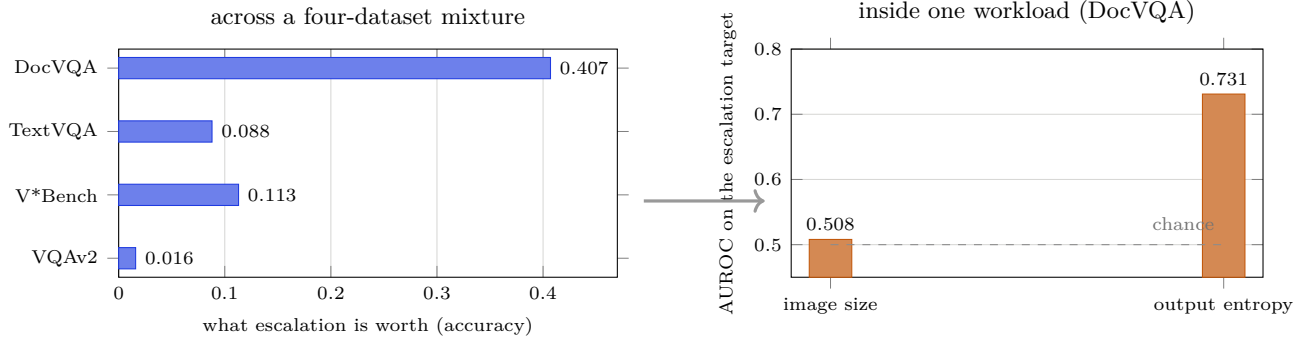


Figure 10: The two regimes, and why a method evaluated on one recommends the wrong signal for the other. **Left:** on a benchmark mixture, what escalation is worth differs by a factor of twenty-five across datasets, so a signal that merely identifies the dataset carries most of the available value without reading anything about a query. Raw image size is such a signal, and it clears the randomisation hull at every operating point there. **Right:** inside a single workload that axis is gone, the same descriptor falls to 0.508 on the escalation target, and only a signal read from the model ranks queries by value. Neither regime is the general case. Bars are measured: full-resolution minus 384 px accuracy per dataset (Fig. 3) and within-DocVQA AUROC (§7.1).

answers only 0.368 of it at the ceiling, so most queries are unanswerable at any rung and there is little for any policy to allocate. What varies is whether *graded* actions add anything on top: on DocVQA the top rung is dear and binary escalation over-serves, so the ladder clears where binary does not; on ChartQA one step is all the corpus supports and the two rules coincide. The claim we make generally is about which signal to read. How much the winning signal is worth, and whether it needs more than two actions, is a property of the corpus and its prices.

**A third lineage, and the exception that separates ranking from allocation.** The workload half of the inversion rested on the SmolVLM family and one run where the model was too weak to allocate anything. LLaVA-OneVision at 0.5B is a third lineage at the same scale, a Qwen2 language model with a SigLIP encoder and anyres tiling, and its rungs are genuinely distinct: 1323, 3051 and 5913 visual tokens at 384, 768 and 1152 px. We ran it on 500 of the same DocVQA pages (Table 13).

**The free descriptor clears the hull here while ranking at chance.** Inside this corpus it predicts the top-rung gain at AUROC 0.518, against output entropy’s 0.731, so it is provably not ordering queries by value; yet its ladder clears at three preferences of four. Cost allocation, not ranking, carries about half of that. This model’s anyres tiling makes the price of escalating a page rise steeply with that page’s size, so allocating budget by size is partly allocating by *cost*, and Proposition 2 shows such a policy can beat a hull built on mean costs without knowing which queries would benefit. §9 measures the share directly by running a policy that reads price and nothing else: it supplies +0.007 of the descriptor’s +0.013 here. The remainder is not accounted for by anything we could measure, and we do not claim it as value ranking.

That distinction was invisible on the SmolVLM family,

| workload and serving model    | entropy ladder   | free descriptor  |
|-------------------------------|------------------|------------------|
| DocVQA, SmolVLM-500M          | 3/4              | 0/4              |
| DocVQA, SmolVLM-256M          | 1/4              | 0/4              |
| DocVQA, Qwen2-VL-2B           | 4/4              | 4/4              |
| DocVQA, LLaVA-OneVision-0.5B  | 3/4              | 4/4              |
| DocVQA, SmolVLM2-2.2B         | 3/4              | 1/4              |
| InfographicVQA, SmolVLM-500M  | 0/4              | 0/4              |
| InfographicVQA, Qwen2-VL-2B   | 4/4 <sup>‡</sup> | 3/4 <sup>‡</sup> |
| ChartQA, SmolVLM-500M         | 4/4              | 1/4              |
| ChartQA, LLaVA-OneVision-0.5B | 0/4              | 3/4              |

Table 13: Operator preferences of four at which each ladder clears the randomisation hull, nominally. The free descriptor clears at three or four preferences on the three pairs whose serving model prices escalation steeply and at no more than one elsewhere, the single ChartQA/SmolVLM-500M clearance not surviving the family-wise correction. On the last row it clears where the model-read signal does not. Table 18 measures how much of each column a policy reading only cost also obtains: at most +0.008 on seven of the eight pairs, so no entry there is displaced by it. <sup>‡</sup>: on InfographicVQA read by Qwen2-VL-2B the signal-free policy reaches +0.054, above both columns, so neither entry on that row is attributable to its signal.

whose patch grid quantises every page to the same handful of token counts, and it narrows what §5.1 is entitled to claim. A free descriptor is a weak value ranker inside every workload we measure, scoring 0.508 to 0.586 on the escalation gain against a model-read signal’s 0.609 to 0.791, and that is what the AUROC column records. It is a better *cost allocator* than value ranker, though only about half of its advantage here is allocation a signal-free policy also obtains, and whether allocating pays at all depends on how steeply the serving model prices its own rungs. On this pair the model-read signal still wins the policy comparison by +0.025 [+0.020, +0.030] at the

loosest preference. It does not win everywhere: §10 reports a sixth pair, crossing both the model and the corpus axes, where the ordering reverses because escalation is barely worth its price and a calibrated rule correctly declines to buy it.

### 6.3 Trying to improve the signal, three times

**A better ranking, again, and again no better policy.** Since our own numbers say the ceiling is the signal, we tried to improve the signal. Entropy and the probe are complementary: a logistic combination of the two reaches 0.805 on the joint target, against 0.751 and 0.761 alone, a gain of +0.054 [−0.010, +0.120].

As a policy it is strictly worse than the probe alone (0.505 accuracy at 181.1 ms, against 0.510 at 154.3 ms) because reading entropy requires finishing the cheap pass, which is precisely the cost the probe existed to avoid. The best ranking and the best policy are different objects here, and the second time we observed this it was not a mistake but a structural feature: a signal’s value to a policy is bounded by what it costs to obtain.

**How far from the oracle?** Not close, and the shape of the gap is itself informative. The oracle escalates only 20% of traffic and is simultaneously more accurate *and* cheaper than always running at full resolution (0.580 at 139.6 ms against 0.530 at 206.0 ms). Measured against the always-cheap baseline, the joint-target probe closes 64% of the accuracy gap but *increases* cost, where the oracle decreases it. No policy we build moves along the direction the oracle occupies.

This is the strongest evidence in the paper that the ceiling is the signal rather than the policy, and it agrees with what the routing literature reports for text: NVIDIA’s prefill-activation router closes 45.6% of its oracle gap [7], and [5] conclude that routing effectiveness is limited by the reliability of the success estimates, not by the routing rule.

**The direction is not domain-general.** Training on detail-limited datasets (DocVQA, V\*Bench) and testing on knowledge-limited ones (VQAv2) gives 0.422; the reverse gives 0.366. Both fall *below* chance: the direction inverts rather than degrading. The 0.760 result holds only because the training fold spans all four datasets. “Would more pixels help” is not one internal quantity but at least two (one for answers present but unreadable, one for answers absent) represented along opposing axes.

## 7 What the probe is actually reading

A cost measurement, not a literature review, exposed this. Pricing escalation per example rather than per configu-

| signal               | pooled       | within one dataset |
|----------------------|--------------|--------------------|
| image size (free)    | 0.748        | 0.534              |
| output entropy       | 0.751        | <b>0.664</b>       |
| pre-generation probe | <b>0.761</b> | 0.519              |

Table 14: Predicting the joint escalation gain. Pooled is the held-out fold of the four-dataset mixture,  $n = 200$ . Within-dataset trains and tests the probe inside a single dataset over 40 resamples, weighted by dataset size, so each fit sees about 210 examples; the single-domain pilot of §7.1 repeats the measurement with nine times the data and reports 0.572 for the probe against 0.663 for entropy.

ration showed the probe systematically escalating *larger* images than entropy does, 1.19 to 1.33× the visual tokens at matched rate. That is worth explaining, and the explanation undermines the result above.

**The flat escalation price was wrong.** Our profiling image had a longest side below 1536 px, so the processor capped it and the “full-resolution” pass spent 320 visual tokens. On the pilot, CROP-free full passes average 424 tokens and exceed the 768 configuration on 44% of examples. Latency is affine in token count ( $100.9 + 0.326v$  ms, worst residual 1.7 ms over four profiled configurations) so the honest per-example price is 238.9 ms against the 206.0 we charged, 16% higher. This is the third measurement error of the same family we have found, after processor upscaling and patch-grid quantisation, and all three come from a configuration name not determining a cost.

Under honest costing the probe’s saving over entropy shrinks from −27.8 [−40.7, −15.0] ms to −11.8 [−27.2, +5.0] ms at a 30% escalation rate: no longer distinguishable from zero. It survives at 20% and 40%.

**Why a better signal picks dearer escalations.** The queries more pixels help are the queries with more pixels to add. A signal that ranks escalation value therefore also selects the expensive tail, and ranking quality partly cancels against cost saving. This is a fourth way a better signal fails a policy, and unlike the first three it is invisible unless cost is measured per query rather than per configuration.

**And the correlation runs deeper than cost.** If image size predicts escalation value, it is a routing signal in its own right, and a free one, read from the file header before any model runs. Table 14 scores it against the two signals the paper has been comparing.

The probe’s margin over a feature costing nothing is +0.013. And the within-dataset column is the result (Fig. 11): with the between-domain axis removed the probe is at chance in every dataset (DocVQA 0.502 [0.483, 0.521], VQAv2 0.468 [0.422, 0.513]) while output entropy retains 0.664. Each of these fits sees only 210 training examples,

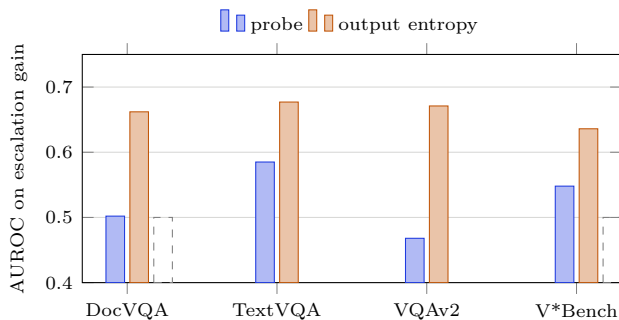


Figure 11: Within a single dataset, trained and tested inside it. The dashed line is chance. The probe’s pooled advantage does not survive removal of the between-domain axis; entropy’s largely does.

|                 | DocVQA     | TextVQA | VQAv2 | V*Bench |
|-----------------|------------|---------|-------|---------|
| test fold       | 30%        | 25%     | 30%   | 15%     |
| probe favours   | <b>50%</b> | 36%     | 12%   | 2%      |
| entropy favours | <b>0%</b>  | 28%     | 46%   | 26%     |

Table 15: Dataset composition of the 50 queries where each signal most wants to escalate relative to the other. Not one of the queries entropy prefers is a DocVQA page.

which is a confound of its own; §7.1 removes it with a pilot built for the purpose.

**What the disagreement looks like.** The statistics above say the probe tracks image size; the queries say what that means. Ranking the held-out fold by each signal and taking the largest rank disagreements, the two sets are almost disjoint by dataset (Table 15).

**The disagreement between the two signals predicts which dataset a query came from at AUROC 0.738**, which is most of what either signal achieves on the escalation target itself. Where they differ, they differ about what kind of image it is.

The individual queries are unambiguous. The four the probe most wants to escalate are all DocVQA, all 448 to 640 visual tokens at full resolution, and all have  $G = 0$ : “What is written in big letters on the top right?” is answered *correctly* at 384 px and the escalation changes nothing. The four entropy most wants to escalate are all VQAv2 photographs at 320 tokens, all wrong at both resolutions, questions like “What is tucked behind the lamp?” where the model is genuinely lost and more pixels do not help.

Read charitably this is the probe doing something reasonable, since documents do repair more often on average. Read as a routing signal it is a category detector: it escalates a correct answer on a large page and declines a hopeless one on a small photograph, and neither decision used anything about the query.

**The text-probe literature saw this coming, and evaluates accordingly.** Having found it, we went back to check whether the work we build on is exposed to the same thing. It is not, and one of them names the mechanism. Moreno Cencerrado et al. [6] train and test their correctness direction *within* each dataset and report cross-dataset transfer separately, and observe that for some sources “the learned direction captures dataset-specific cues correlated with correctness rather than correctness itself”. Lugoloobi et al. [5] likewise average AUROC over their benchmarks rather than pooling examples across them. Neither pools; we did.

Two of their observations also validate the test we use here. Cencerrado et al. report that probe performance on GSM8K “plateaus near random chance regardless of dataset size”, distinguishing an absent signal from an under-sampled one by exactly the learning-curve argument, and they find a non-linear classifier on external embeddings generalising *worse* than a linear probe on internal states, the opposite of what a capacity explanation would predict. The correction here is therefore to our evaluation protocol, not to their finding.

**This explains a result we had already reported and misread.** We found earlier that the probe’s direction inverts across domains (trained on detail-limited data and tested on knowledge-limited data it scores 0.422, and the reverse 0.366, both below chance) and read it as a transfer limitation calling for a domain-representative calibration set. The correct reading is stronger: a direction that carries no within-domain signal and inverts between domains is a domain classifier, and its pooled AUROC measures how separable the mixture is, not how well the model knows its own needs. Both facts were in our data; only the second explains them.

## 7.1 A single-domain pilot, and what it settles

The within-domain fit above gets 210 training examples where the pooled fit gets 600, so part of the collapse could be sample size rather than absent signal. We said so, and then collected the pilot that decides it: 1200 DocVQA pages, enough to fit and test inside one domain.

The decisive form is a learning curve. If the probe were merely starved, AUROC would climb toward the pooled figure as training data grows. Neither extreme is what happens (Fig. 12).

The probe does improve, by +0.050 as training data grows ninefold, so the collapse is not purely starvation. But it plateaus far below output entropy, which is fitted on nothing. **The signal is genuinely weaker inside a domain than the pooled figure suggests**, and the confound conclusion stands in this corrected form rather than as “at chance”.

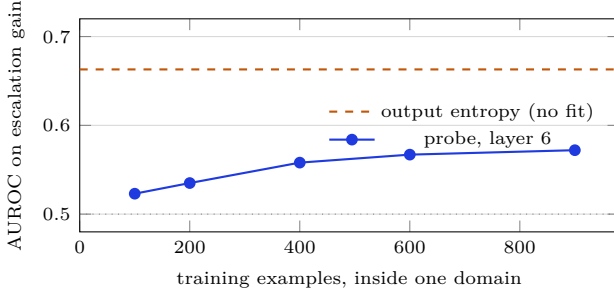


Figure 12: Probe AUROC inside DocVQA against training size, 30 resamples with a fixed 300-example held-out set. The dotted line is chance. More data helps and does not close the gap: at 900 examples, more than the pooled fit that produced 0.761, the probe reaches 0.572.

#### And the depth that works does not transfer either.

We selected layer 6 because AUROC saturates there on the mixture, and the entire read-cost argument of §5 rests on stopping a sixth of the way through the language model. Inside one domain the signal keeps improving with depth:

| layer | 1     | 3     | 6     | 12    | 20    | 32           |
|-------|-------|-------|-------|-------|-------|--------------|
| AUROC | 0.531 | 0.546 | 0.572 | 0.579 | 0.618 | <b>0.625</b> |

The best depth is the last one. That costs the probe its cheap read: Eq. 7 gives 20.0 ms at  $\ell = 6$  and 59.1 ms at  $\ell = 32$ , so the saving per escalated query falls from 106.9 to 67.8 ms.

That residual 67.8 ms is exactly the decode time, because a full-prefill probe still skips generation, and it is the same floor the cache analysis of §3.2 arrived at from an unrelated direction. The probe’s advantage narrows to one mechanism, stated twice: reading the model without making it speak.

**An independent corroboration of the corrected depth.** GlimpsePrune [4] reads its learned glimpse token at layer  $K = \frac{2}{3}L$ , chosen as the point where it has “aggregated sufficiently clear importance information” at a favourable overhead trade-off. That is where our within-domain sweep becomes informative (0.618 at layer 20 of 32) and far from the layer 6 the mixture selected. Two groups reading different internal signals for different purposes land on the same region of the network, which is weak evidence but it points the same way as §7.1: the shallow saturation was the artefact, not the deep one.

**What survives.** The mixture result is real, and mixtures are real deployments: a router serving documents and photographs through one endpoint faces exactly the pooled distribution, and there a free image-size feature is not merely nearly as good but, built into a policy and priced, indistinguishable from the probe within 0.010 at every preference (Table 8). The negative results are untouched, since none of them rests on the probe working.

And the paper’s thesis survives inverted: we set out to show that a better signal is not a better router, and the sharpest instance turns out to be our own signal, whose advantage was an artefact of the evaluation mixture.

**What does not.** The claim that a 500M VLM encodes its need for more pixels in a linearly readable direction. On the evidence here it mostly encodes what kind of image it is looking at, which correlates with that need across a benchmark mixture and much less within one. §7.1 is the test we said this needed, run on a corpus built for it, and it does not rescue the claim: given nine times the training data the direction reaches 0.572 where the model’s own output entropy, fitted on nothing, reaches 0.663. The honest statement is that the direction carries some within-domain signal and less than the cheaper alternative.

## 8 The escalation rate is not a free parameter

Every policy above picks a rate and thresholds at that quantile. That is a tuning exercise, and nothing connects the chosen rate to Eq. 1, the cost function the oracle is already labelled under. Kotte’s UCCI [11] closes the same gap for text cascades (calibrate the signal to a probability, then let the cost model choose the threshold) and its transposition here is short, but it needs one correction.

**The rule.** Escalating  $x$  changes expected cost by

$$\Delta J(x) = -w_e \mathbb{E}[G \mid x] + \lambda_t \Delta t, \quad (10)$$

so the optimal action is to escalate exactly when

$$\mathbb{E}[G \mid x] > \tau = \frac{\Delta t}{V}, \quad V = \frac{w_e}{\lambda_t}, \quad (11)$$

where  $V$  is the only knob: the latency an operator will spend to buy one additional correct answer, in milliseconds. Following [14], we state the trade-off in the units of the outcome rather than as a weight per millisecond, because the latter is not readable across devices. Estimating  $\mathbb{E}[G \mid x]$  from the probe score by isotonic regression (monotone, non-parametric, one pass over cached activations) gives Algorithm 1.

#### Why isotonic, and what actually matters about it.

We take isotonic regression from [11], and borrowing a component is how a paper inherits someone else’s assumptions. The settings differ where calibration is sensitive: their target is a probability with tens of thousands of calibration points, ours a signed gain with a few hundred. Three families thresholded at the *same* break-even, so any difference is the calibrator’s magnitudes rather than a different operating point (Table 16).

The sigmoid ranks marginally *best* and calibrates worst, at nearly twice the error, and its policy is significantly less

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**Algorithm 1** Calibrated gain-optimal escalation

---

**fit**

```

1 read  $h^{(\ell)}$  at the last prompt position,  $\ell = 6$ 
2  $G_i \leftarrow \mathbf{1}[\neg y_i^{\text{cheap}} \wedge y_i^{\text{full}}] - \mathbf{1}[y_i^{\text{cheap}} \wedge \neg y_i^{\text{full}}]$ 
3  $w \leftarrow \mu_+ - \mu_-$  over  $\{G_i > 0\}$  vs. the rest
4  $s_i \leftarrow (h_i - \bar{\mu})^\top w / \|w\|$ 
5  $\hat{g} \leftarrow 2 \text{isotonic}(s, (G+1)/2) - 1$ 
serve  $x$ , given  $V$ 
6  $\Delta t \leftarrow t_{\text{probe}} + t_{\text{full}} - t_{\text{cheap}}$ 
7 if  $\hat{g}(s(x)) > \Delta t / V$  then abandon the prefill, run full
8 else finish the cheap pass and answer

```

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| calibrator      | AUROC        | calib. error | accuracy     |
|-----------------|--------------|--------------|--------------|
| isotonic        | 0.660        | 0.059        | <b>0.736</b> |
| Platt sigmoid   | <b>0.667</b> | 0.098        | 0.713        |
| equal-mass bins | 0.656        | <b>0.051</b> | 0.733        |

Table 16: Calibrator families on 1200 DocVQA pages, 30 resamples, all thresholded at  $\tau = 0.243$ . Paired against isotonic, Platt loses 0.022  $[-0.026, -0.019]$  accuracy and equal-mass bins differ by  $-0.003 [-0.006, 0.000]$ .

accurate. That is the paper’s recurring pattern arriving in a third place: ranking quality does not transfer, this time because a rule that thresholds a magnitude is broken by a miscalibrated magnitude in a way a rank threshold would never notice. A parametric squash reports gains the operator did not ask for, and the rule lands somewhere other than the point  $V$  specifies.

Equal-mass bins, meanwhile, are indistinguishable from isotonic. **The load-bearing property is being non-parametric, not being isotonic**, and we say so rather than defend a component we happened to inherit. Isotonic remains the better default because monotonicity is free information that binning discards, but nothing here rests on it.

**When the rule is optimal, and how it relates to UCCI’s theorem.** Proposition 1 below is a direct transposition of the optimality argument of [11] to a different calibration target; we state it to fix notation and to make visible exactly which assumption separates the two rules, not as a contribution in itself. Let  $s(x)$  be any routing score and let  $\Pi_s$  be the policies measurable with respect to it, which is what a deployed router is: it sees the score, not the outcome.

**Proposition 1.** *Under the cost of Eq. 1 with  $w_e, \lambda_t > 0$ , and given the calibrated gain  $\hat{g}(s) = \mathbb{E}[G \mid s]$ , the policy*

$$\pi^*(x) = \mathbf{1}[\hat{g}(s(x)) > \Delta t / V], \quad V = w_e / \lambda_t,$$

*minimises expected cost over  $\Pi_s$ . If  $\hat{g}$  is non-decreasing, it is a threshold on  $s$  itself.*

*Proof.* Write  $C(\pi) = C_0 + \mathbb{E}[\pi(x) \Delta J(x)]$ , where  $C_0$  is the cost of answering cheap on every query and  $\Delta J$  is Eq. 10.

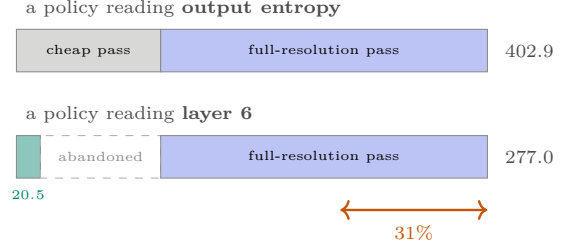


Figure 13: Why a signal’s read cost changes what it is worth, on an escalated query. Reading output entropy requires finishing the cheap pass it is meant to price, so escalation adds a full-resolution pass to one already paid for. Reading the residual stream at layer 6 produces the score after 20.5ms and abandons the rest, which is 277.0ms against 402.9, a measured 31%. Both numbers come from physically truncating the decoder to six layers rather than from arithmetic. The consequence is not only a smaller bill: at the same operator preference the cheaper signal faces half the break-even gain, so **it is entitled to escalate on weaker evidence**. What standard serving stacks do not expose is the control to abandon a prefill per query.

The expectation is linear in  $\pi$  and the constraint set is  $\pi(x) \in \{0, 1\}$ , so it is minimised pointwise by escalating exactly where  $\Delta J(x) < 0$ . For  $\pi \in \Pi_s$  the tower property gives  $\mathbb{E}[\Delta J \mid s] = -w_e \mathbb{E}[G \mid s] + \lambda_t \Delta t$ , so the pointwise test becomes  $\hat{g}(s) > \lambda_t \Delta t / w_e$ . Monotone  $\hat{g}$  makes the sublevel set an interval in  $s$ .  $\square$

Isotonic regression returns a non-decreasing  $\hat{g}$  by construction, so Algorithm 1 realises  $\pi^*$  whenever the calibration is accurate, which Fig. 14 measures rather than assumes. And the relation to Kotte’s Theorem 1 [11] is exact: under their assumption that the escalated pass attains a fixed accuracy  $\gamma$  independent of which queries are sent,  $\mathbb{E}[G \mid x] = \gamma - (1 - p_{\text{err}}(x))$ , which is increasing in the weak model’s error probability, so their rule and ours coincide. **The two rules agree exactly when escalation cannot make things worse, and Table 17 is what their difference costs when it can.**

**The break-even is signal-dependent, and that is the probe’s second advantage.** Line 6 differences Eq. 9. A policy reading entropy must finish the cheap pass before deciding, so escalation adds the whole full-resolution pass,  $\Delta t = 206.0$ ms. A policy reading the probe abandons the cheap pass it never completed, so  $\Delta t = 20.3 + 206.0 - 123.4 = 102.9$ ms, exactly half. At the same operator preference  $V$ , the probe’s break-even gain is half entropy’s: **a cheaper signal is entitled to escalate on weaker evidence**, and the saving compounds rather than merely lowering the bill for a fixed rate.

| rule        | calibration target      | escalates  | accuracy | latency      |
|-------------|-------------------------|------------|----------|--------------|
| UCCI [11]   | $P(\text{cheap wrong})$ | 54%        | 0.520    | 178.5        |
| <b>ours</b> | $\mathbb{E}[G \mid x]$  | <b>34%</b> | 0.520    | <b>157.9</b> |

Table 17: Both rules read the same probe score at  $V = 800$  ms per correct answer and differ only in what they calibrate. Paired over the held-out fold ( $n = 200$ ), the latency difference is  $-20.6$   $[-26.8, -14.9]$  ms and the accuracy difference  $0.000$   $[-0.025, 0.025]$ . A two one-sided test establishes equivalence only against a margin of  $0.020$ , so a loss of up to two accuracy points is not excluded; within that margin, the error-probability target costs 59% relatively more compute for no measured accuracy.

**Calibrating the wrong probability.** Condition (ii) of UCCI’s Theorem 1 is that the large model attains a fixed accuracy on the test distribution, invariant to which queries are escalated. Under it the escalation value of a query is determined by the weak model’s error probability, so calibrating  $P(\text{cheap wrong})$  suffices. **Kotte identifies that condition as the most restrictive of the three and validates it**, reporting a strong model at micro-F1 0.928 on escalated queries against 0.932 on the full test set, a difference of 0.004. It is therefore not a convenience assumption but a checked one, and what we report is that the check does not transfer. Visual escalation violates it: the second pass repairs 21% of queries and *damages* 4%, so it is a second random outcome rather than a fixed accuracy, and the quantity to calibrate changes with it. The quantity to calibrate is the signed gain, and the difference is not academic (Table 17).

**This is an equivalence of aggregate accuracy, not of answers.** The two rules escalate 34% and 54% of queries and disagree on 20% of them, so they return different answers on one query in five; what the test above supports is that those differences cancel to within two accuracy points. The bound is loose because the held-out fold is 200 examples, and a tighter one would need a larger pilot rather than a better test [RESULT NEEDED].

The failure is legible. A confidently wrong cheap pass has high error probability and near-zero expected gain, so UCCI escalates it and the gain rule does not, the same conditional-versus-joint error of §5.1, displaced from the training target to the calibration target. Fitting the gain directly is well calibrated: binned over the held-out fold, predicted and realised gain agree to 0.029 (Fig. 14), against a correctness ECE of 0.027 for the same signal, so the correction costs nothing in calibration quality.

**The contrast on a second sub-billion model.** Table 17 is one model, so we repeated it on SmolVLM-256M over the same images, the same signal and the same preference. The direction replicates and the failure mode is sharper: calibrating the weak pass’s error probability sends *every* query to full resolution, where calibrating the signed gain sends 58%, and the gain rule is

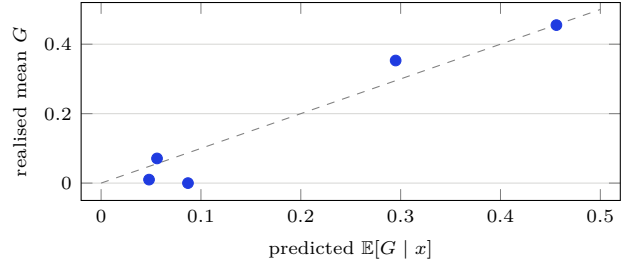


Figure 14: Reliability of the gain calibration on the held-out fold, binned by predicted gain. The dashed line is perfect calibration. A rule that thresholds a *magnitude* is only as good as the magnitude, which a ranking metric would not detect.

43.2  $[36.5, 50.4]$  ms per query cheaper at an accuracy difference of  $-0.030$   $[-0.065, +0.005]$  that spans zero. On the 500M model the same comparison gives 20.6  $[14.9, 26.8]$  ms and  $0.000$   $[-0.025, +0.025]$ .

The reason the smaller model makes it worse is the reason the correction exists. A weaker cheap pass is wrong more often, so  $P(\text{cheap wrong})$  rises for almost every query while the *gain* from escalating does not, since more of those failures are unrecoverable at any resolution. A rule thresholding the first therefore saturates, and a rule thresholding the second does not. **The correction matters more, not less, as the serving model gets weaker**, which is the direction this paper’s scale argument runs in.

**Does removing the knob cost anything?** It should not, if the rule is right. Sweeping  $V$  over 100 to 3200 ms per correct answer places six operating points on the accuracy-latency plane without any rate ever being chosen; five of the six are undominated by the tuned sweep of Fig. 7 evaluated on the same fold (Fig. 15), and 78% remain undominated across the 200 resplits. An untuned rule reaches a tuned frontier, which is the useful form of this result: the rate was never information, only a stand-in for a preference the operator already has.

**Where the rule refuses.** At  $V \leq 200$  the break-even of Eq. 11 exceeds any attainable gain and the rule escalates nothing, returning the always-cheap policy. That is the correct answer, not a degenerate one: by Eq. 10 no expected gain on this pilot repays 102.9 ms when a correct answer is worth 200. A tuned rate cannot express this, which is the failure mode reported for our earlier threshold tuners, they discovered “never escalate” on one fold and had no way to say whether that was a preference or an artefact. Where a floor on escalation is required instead, [12]’s recall-controlled gates give one that cannot degenerate by construction.

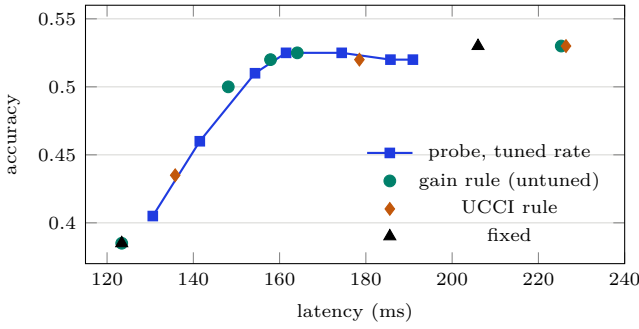


Figure 15: Untuned decision rules against the tuned sweep. The gain rule tracks the swept frontier without a rate parameter; the UCCI rule, calibrating correctness instead, sits inside it and collapses to always-escalate once  $V$  is large.

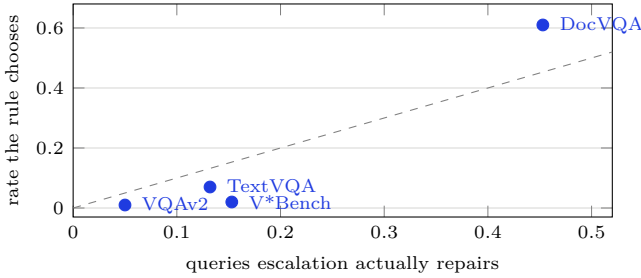


Figure 16: One value of  $V$ , applied unchanged to four domains. The rule escalates 61% of DocVQA and 1% of VQAv2, tracking how much escalation actually repairs there ( $r = 0.974$ ). The dashed line is not a target, escalating every repairable query is not optimal, but it fixes the scale.

## 8.1 The rule transfers where the signal does not

Every number above uses the probe, which §7 discredits within a domain. So we re-ran the whole comparison on output entropy, the signal that does work within a domain, fitted and tested inside each dataset over 40 resamples, with escalation priced per example. If the rule is a property of the method rather than of the signal, it should survive.

It does, and the confound makes it more useful rather than less. Escalation value is domain-determined, so a fixed rate is wrong in every domain but the one it was tuned on. A rule calibrated on the observed gain distribution picks its own rate wherever it is fitted, from one unchanged operator preference (Fig. 16).

Fixing one policy and applying it to all four domains, weighted by dataset size, the rule reaches 0.489 accuracy at 181.6 ms where a tuned 30% rate reaches 0.464 at 195.1 ms: more accurate *and* cheaper. Tuned rates at 40% and 50% are dominated as well. Only the two lowest tuned rates survive on the front, at the cautious end where escalating little is right everywhere.

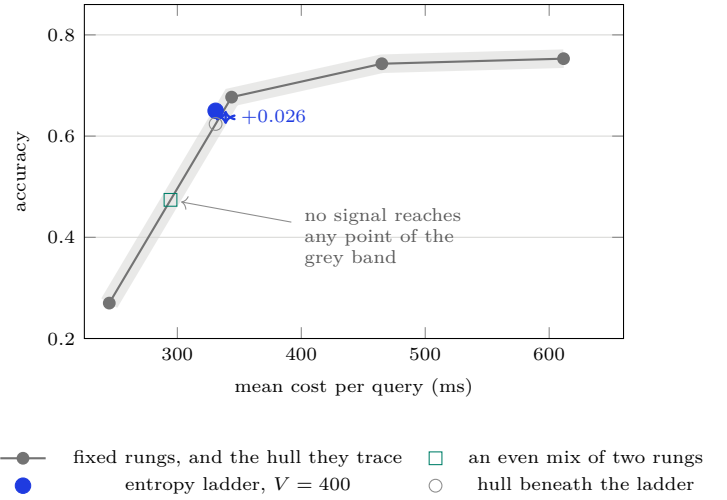


Figure 17: What a hull gap is, on DocVQA read by SmolVLM-500M, one held-out fold of 300 pages (seed 0, regenerated by `make_hull_figure.py`). Randomising between two fixed rungs at rates  $p$  and  $1 - p$  lands on the segment joining them, so the grey band is reachable with no signal at all: it is the floor an adaptive policy must clear, and the endpoints this literature compares against are its two least interesting points. The ladder sits +0.026 above it here; averaged over thirty folds the gap is +0.041 (Table 13), and the two differ because the hull is curved, so the mean of per-fold gaps is not the gap between averaged points. Costs are the modelled per-example prices of §3.2. Of a gap this size, at most +0.003 is reachable by a policy reading price alone (Table 18); Proposition 2 states the general form.

## 9 What a hull gap is worth, and what it is not

The comparator of §6 decides every claim in this paper, so it deserves a statement of what clearing it proves, a statement of what it does not, and a measurement of how permissive it is. This section supplies the three in that order, and the third costs us a corpus-model pair.

**What clearing the hull certifies.** The conclusion drawn from a hull gap is a claim about information, so it needs a statement rather than an intuition. Write  $c_i(x)$  for the cost of serving query  $x$  at configuration  $i$  and  $y_i(x) \in \{0, 1\}$  for its correctness there, and let a policy  $\pi$  be measurable with respect to a signal  $s$ , meaning it sees  $s(x)$  and nothing else.

**Proposition 2.** *If  $s$  is independent of the joint vector  $(c(x), y(x))$ , then the operating point  $(\mathbb{E}[\text{cost}], \mathbb{E}[\text{accuracy}])$  of any  $s$ -measurable policy lies in the convex hull of the fixed points  $(\mathbb{E}[c_i], \mathbb{E}[y_i])$ .*

*Proof.* Condition on  $s$ . Writing  $\pi_i(s)$  for the probability the policy serves configuration  $i$ , independence gives

$\mathbb{E}[y_\pi | s] = \sum_i \pi_i(s) \mathbb{E}[y_i]$  and likewise for the cost, since conditioning on  $s$  changes neither marginal. Taking the expectation over  $s$  yields  $\sum_i \mathbb{E}[\pi_i(s)] \mathbb{E}[y_i]$  with weights that are non-negative and sum to one, which is a convex combination of the fixed points.  $\square$

The contrapositive is the comparator this paper uses: a policy strictly above the hull certifies that its signal is *not* independent of  $(c, y)$ . Two things follow, and the second is why the proposition is worth stating.

First, the hull is the right floor and a single fixed end-point is not. Comparing an adaptive policy to always-cheap or always-full establishes nothing, because randomising between them already reaches every point of the segment between them at no informational cost. That comparison is standard in text routing, where [5] plot their cascades against it, and absent from the visual escalation evaluations we survey.

Second, and less comfortably, the certificate is about the *pair*  $(c, y)$ , not about correctness alone. A signal carrying no information about which queries escalation would repair, but some about what escalating them costs, clears the hull by allocating budget rather than by ranking value.

That is not a hypothetical, and we state its status carefully. We wrote this proposition *after* observing a free descriptor clear the hull on one serving model while scoring at chance on the escalation target (§10), so what follows is a post-hoc explanation, not a forecast. What makes it more than a restatement is that it implied a quantity we had not measured and which could have come out flat: if cost is the channel, the descriptor must correlate with per-page escalation cost on the model where it clears and not on the model where it does not.

| corpus, serving model  | $\rho(\text{size}, \Delta t)$ | spread | free       | entropy    |
|------------------------|-------------------------------|--------|------------|------------|
| InfoVQA, SmolVLM-500M  | +0.008                        | 1.6×   | 0/4        | 1/4        |
| DocVQA, SmolVLM-256M   | +0.142                        | 2.0×   | 0/4        | 1/4        |
| DocVQA, SmolVLM-500M   | +0.083                        | 2.3×   | 0/4        | 3/4        |
| DocVQA, SmolVLM2-2.2B  | n/a                           | 5.4×   | 1/4        | 3/4        |
| InfoVQA, Qwen2-VL-2B   | n/a                           | 16.3×  | 3/4        | 4/4        |
| DocVQA, Qwen2-VL-2B    | +0.342                        | 41.1×  | 4/4        | 4/4        |
| ChartQA, LLaVA-OV-0.5B | +0.151                        | 53.4×  | <b>3/4</b> | <b>0/4</b> |
| DocVQA, LLaVA-OV-0.5B  | +0.336                        | 77.4×  | 4/4        | 3/4        |
| DocVQA, InternVL3-1B   | −0.099                        | 1.00×  | excluded   |            |
| ChartQA, SmolVLM-500M  | +0.130                        | 1.26×  | excluded   |            |

Operator preferences of four at which each ladder clears the hull, ordered by how steeply the serving model prices escalation. The last two rows are excluded because configurations that barely differ in cost make a hull gap arbitrarily sensitive to a millisecond. The two rows marked n/a are pairs we added to interpolate the price axis and for which we did not extract the size-cost correlation, since the ordering variable is the spread. That criterion is a property of the pair and not of the result, so it applies wherever a hull gap is read: the ChartQA/SmolVLM-500M entries in Table 13 are reported for completeness and are not used as evidence anywhere, including where they favour us.

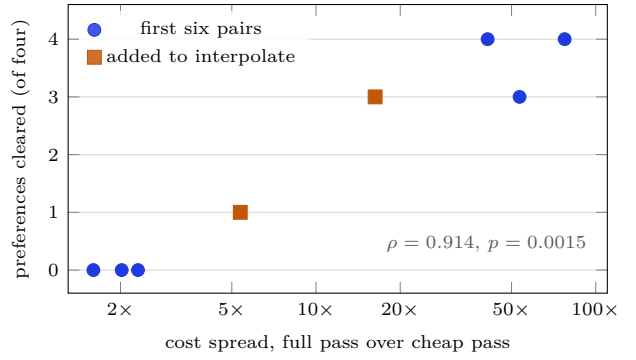


Figure 18: How often the free descriptor’s policy clears the randomisation hull, against how steeply its serving model prices escalation. Each point is one corpus-model pair and the vertical axis counts operator preferences of the four we sweep. Our first six points left an empty decade between 2.3×

**What the contrapositive does not give.** Exact independence is a convenience of the statement, not a description of any real signal, so the certificate is qualitative: it separates signals carrying *some* information about  $(c, y)$  from those carrying none, and almost every signal is in the first class. The size of a hull gap has no direct informational reading, because it depends on the conditional variance of  $(c_i, y_i)$  given the signal and on where the operating budget falls between the fixed points, so the same signal can show a large gap at a tight budget and none at a loose one, as Table 7 does. We therefore use the gap as a comparator between policies at matched budget, which is what every table reports, and never as a measure of how much a signal knows. What we can put a scale on is the part of a gap that needs no signal at all, and Table 18 measures it per pair.

**The correlation is not what separates them, the price is.** We first read the split as the descriptor tracking cost, and one row refutes that: the descriptor clears at  $\rho = +0.151$  on ChartQA/LLaVA-OneVision where it clears nothing at a slightly higher +0.142 on DocVQA/SmolVLM-256M. What orders the pairs is the cost spread, and over these eight it does so monotonically: the descriptor clears at 0, 0, 0, 1, 3, 4, 3 and 4 preferences of four as the spread rises from 1.6×

withdrawal concerns whether the clearance is attributable to the *signal*, and only the former enters here. Two of the eight, at  $5.4\times$  and  $16.3\times$ , were added specifically to fill the gap our first six left between  $2.3\times$  and  $41.1\times$ , since a threshold placed inside a void is not a measurement. The relationship is still one we found rather than predicted, and we label it as post-hoc.

**And the ordering claim has a counterexample, in the row that crosses both axes.** On ChartQA read by LLaVA-OneVision, the free descriptor clears at three preferences and the model-read signal at none. The reason is not that the descriptor ranks better: it scores 0.586 on the escalation gain against entropy’s 0.609. It is that escalation there repairs 12% of queries and damages 5%, a net 7% for a pass costing  $53\times$  the cheap one, so a calibrated rule reading the gain correctly declines to escalate at every preference we sweep and sits on the cheap vertex, where its gap is zero by construction.

**This is a limitation of the comparator, not a virtue of the descriptor.** The hull is built from the mean cost of each fixed configuration. A policy whose signal correlates with *per-example* cost can therefore beat it by escalating the cheap instances, without any information about which instances would benefit. Proposition 2 says as much, since its independence condition is over the joint vector and not over correctness alone; what the crossed cell adds is that the loophole is large enough to reverse an ordering we had stated generally. How large it is in absolute terms is an empirical question rather than a corollary, and Table 18 answers it: at most  $+0.008$  on seven of the eight pairs, and  $+0.054$  on the eighth, where it exceeds both signals. Where it is small it reverses an ordering only because the ordering it reverses is itself worth about that much. We therefore retract that ordering as a general claim, keep the hull as the floor it is, and read it alongside the target-specific AUROC throughout: **a hull gap is evidence of information about the cost-correctness pair, and only a measurement against the target says which of the two a policy is using.**

**How much of a gap needs no signal at all.** The loophole above is a statement about the comparator and it deserves a number rather than a description. We measure it directly: a policy that escalates the queries *cheapest to escalate* up to a budget, reading the per-example price and nothing else, swept over rates and scored against the same hull (Table 18). A server can estimate that price before deciding, since these models charge by visual-token count, so this is a deployable policy and not only an analytic device.

**The loophole is real but small, and the naive version of it does not work at all.** A binary cost-only policy sits *below* the hull on seven of the eight pairs, because the queries that are cheap to escalate are also the ones escalation helps least, and buying them at the top rung’s mean price is a bad trade. The graded policy, which spends the same budget on more queries at smaller steps,

| corpus, serving model  | spread       | cost-only |               | our ladders |        |
|------------------------|--------------|-----------|---------------|-------------|--------|
|                        |              | binary    | graded        | entropy     | free   |
| InfoVQA, SmolVLM-500M  | $1.6\times$  | −0.004    | −0.011        | +0.006      | −0.004 |
| DocVQA, SmolVLM-256M   | $2.0\times$  | −0.028    | +0.004        | +0.023      | −0.001 |
| DocVQA, SmolVLM-500M   | $2.3\times$  | −0.049    | +0.003        | +0.041      | +0.000 |
| DocVQA, SmolVLM2-2.2B  | $5.4\times$  | −0.036    | +0.003        | +0.040      | +0.005 |
| InfoVQA, Qwen2-VL-2B   | $16.3\times$ | +0.030    | <b>+0.054</b> | +0.043      | +0.018 |
| DocVQA, Qwen2-VL-2B    | $41.1\times$ | −0.020    | +0.008        | +0.075      | +0.018 |
| ChartQA, LLaVA-OV-0.5B | $53.4\times$ | −0.008    | +0.001        | −0.000      | +0.010 |
| DocVQA, LLaVA-OV-0.5B  | $77.4\times$ | −0.011    | +0.007        | +0.038      | +0.013 |

Table 18: The largest gap reachable while reading *only* per-example price, in the binary action space and in the graded one, against the largest gap our own two ladders reach. Thirty resampled folds; the cost-only policies order by predicted visual-token count, so no signal about the query enters. A reported gap is attributable to its signal only above the graded column. Rows are ordered by cost spread, the ratio of mean full-pass to mean cheap-pass latency. The bold row is the one pair where a signal-free policy beats both of our signals.

does clear, but on seven pairs never by more than  $+0.008$ . Every clearance we report on those seven survives that bar, the entropy ladder’s  $+0.041$ ,  $+0.023$ ,  $+0.075$ ,  $+0.040$  and  $+0.038$  against slacks of  $+0.003$ ,  $+0.004$ ,  $+0.008$ ,  $+0.003$  and  $+0.007$ . **On those pairs the randomisation hull is a nearly tight floor**, measured rather than assumed, which is what makes it usable as the comparator we ask this literature to adopt.

**The eighth pair is the exception and we do not explain it away.** On InfographicVQA read by Qwen2-VL-2B a graded cost-only policy reaches  $+0.054$ , above both our entropy ladder’s  $+0.043$  and the free descriptor’s  $+0.018$ , so *neither* clearance on that pair is attributable to its signal and we withdraw both. It is also the only pair where the binary policy clears. What distinguishes it is that escalation there is both expensive and unusually valuable: this is the corpus where the same model keeps gaining past our highest rung (§10), so a policy that merely buys more escalations for the money captures a great deal without knowing which ones to buy. The lesson is not that the comparator is unusable but that its slack is a per-pair quantity and has to be measured per pair, which costs one sweep.

**Two comparators, and the difference is what a server can know.** Our first version of this experiment ordered queries by their *measured* escalation latency and reported slacks up to  $+0.053$ , large enough to withdraw two of our own results. We first read that as an artefact of timing: each pass in the single-domain pilots is timed once (§13), and on the SmolVLM ladder a fifth of first-step latencies come out non-positive, which looked like noise rather than a free upgrade. **Re-timing says otherwise and we withdraw that explanation.** On a subsample re-run with three repeats after a discarded warmup (Table 19), the escalation step’s spread across examples is 4.9 times its measurement spread, and the non-positive steps

| rung       | mean  | within-example | across examples |
|------------|-------|----------------|-----------------|
| 384 px     | 459.5 | 28.4           | 328.7           |
| 768 px     | 590.3 | 42.7           | 421.1           |
| 1152 px    | 717.5 | 44.3           | 415.7           |
| full       | 968.4 | 48.9           | 517.7           |
| first step | 79.1  | 53.7           | 261.7           |

Table 19: Latency in milliseconds on a re-timed subsample of 150 DocVQA pages, three repeats after one discarded warmup. The within-example column is the measurement spread a policy faces; the last is the variation it ranks on. Their ratio is 4.9 on the escalation step, so per-example price is mostly real rather than measurement error. Absolute latencies exceed those of Table 2 because this run is an hour of sustained load on a laptop GPU; the ratios, which are what the argument uses, are unaffected.

survive averaging, 20.7% from one timing and 19.3% from three, while *none* of them costs zero extra visual tokens. They are neither noise nor patch-grid quantisation: for a fifth of pages this serving stack is not monotone in visual-token count, and the extra tokens are absorbed by costs that do not scale with them.

What separated the two measurements was therefore not noise but *information*. A policy ordering by measured latency knows each query’s realised escalation price before deciding, which no server can, since the price of a pass is not available until the pass has run. Ordering by visual-token count, a deterministic function of the image and the processor, is the deployable version of the same idea and is the one Table 18 reports. **Read as a bound on what a deployable signal-free policy achieves, the slack is at most +0.008 on seven pairs; read as a bound on what an oracle over cost achieves, it reaches +0.053, and two of our clearances would not survive that stronger floor.** We report the first because it is the comparator an operator faces, and the second here because a reader who prefers the stronger floor is entitled to the number [À VÉRIFIER: the oracle figures come from the ordering we retracted and were not recomputed on the re-timed subsample].

**Cost allocation is a partial mechanism, and this puts a share on it.** §5.1 read the free descriptor’s clearance on steeply priced models as budget allocation rather than value ranking. The graded cost-only policy is that account in its purest form, and it accounts for 46% of the descriptor’s gap on Qwen2-VL-2B (+0.008 of +0.018) and 54% on LLaVA-OneVision (+0.007 of +0.013), while on the two SmolVLM pairs both quantities are indistinguishable from zero together. Across the eight pairs the slack and the descriptor’s gap rank alike, which is consistent with the account rather than evidence for it [À VÉRIFIER: eight points, post-hoc]. What it does not explain is the crossed cell, where the descriptor reaches +0.010 and cost allocation supplies +0.001 of it. **Roughly half of the free descriptor’s advantage on steeply priced**

| corpus, serving model  | whether | how far      | spread | length |
|------------------------|---------|--------------|--------|--------|
| DocVQA, SmolVLM-500M   | 0.508   | 0.509        | 0.521  | 0.507  |
| DocVQA, SmolVLM-256M   | 0.509   | 0.508        | 0.519  | 0.512  |
| DocVQA, SmolVLM2-2.2B  | 0.516   | 0.491        | 0.508  | 0.513  |
| InfoVQA, SmolVLM-500M  | 0.508   | <b>0.692</b> | 0.504  | 0.516  |
| InfoVQA, Qwen2-VL-2B   | 0.568   | <b>0.640</b> | 0.568  | 0.540  |
| DocVQA, Qwen2-VL-2B    | 0.520   | 0.478        | 0.513  | 0.510  |
| ChartQA, LLaVA-OV-0.5B | 0.586   | 0.472        | 0.545  | 0.600  |
| DocVQA, LLaVA-OV-0.5B  | 0.518   | 0.465        | 0.513  | 0.531  |

Table 20: AUROC of raw image size against four targets. *Whether*: does the top rung repair this query, the target scored throughout the paper. *How far*: among repaired queries, is more than one rung needed, which is the question a graded ladder faces once it has decided to spend. *Spread*: is the per-rung gain dispersed. *Length*: do the cheap and top answers differ in length, the shallow-cue hypothesis. Bold marks the two cells whose interval excludes chance, and they are not the pairs where the descriptor’s policy clears. Intervals are percentile bootstrap over examples; the how-far column is defined only on repaired queries, which is between 59 and 596 of them.

**models is budget allocation available without any signal; the remainder is unexplained and we do not claim it as value ranking**, since the descriptor’s AUROC on the escalation target is 0.518 to 0.586. We test four accounts of the residual below: one separates only on the pairs where the descriptor’s policy does not clear, one is undecided at our sample, and two we set aside.

**Four accounts of the residual, and what each is worth.** Cost allocation leaves about half of the descriptor’s gap unexplained on the steeply priced pairs, so we looked for the rest rather than leaving it as an admission. Each hypothesis below predicts that image size separates a target the top-rung gain cannot see, and each was scored on the same records (Table 20). One separates, on the wrong pairs; two do not; and the fourth cannot be decided with the sample the question leaves us.

*Rung selection* was the most promising, because the descriptor’s *binary* policies sit at +0.000 on every pair while its graded ladders are what clear, so whatever it supplies is not a whether-signal. It does separate, but not where it would have to. On the two InfographicVQA pairs the how-far AUROC is 0.692 [0.610, 0.777] and 0.640 [0.582, 0.711], intervals excluding chance; on those pairs the descriptor’s policy clears at 0/4 and at 3/4 of which none is attributable. On the three pairs where its policy does clear, the AUROC is 0.478 [0.390, 0.546], 0.472 [0.351, 0.608] and 0.465 [0.407, 0.530]. **Those three intervals span chance, so the account is undetected there rather than refuted**, and the distinction matters: the target is defined only on repaired queries, of which those pairs supply 161, 59 and 131. We report the account as untested at the sample the question leaves us [RESULT NEEDED: a corpus large enough to give several hundred repaired

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**Algorithm 2** Pricing a policy against the no-signal floor

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**given** configurations  $1 \dots R$  in increasing cost,  
a policy  $\pi$ , and a held-out fold  $T$   
**hull**  
1  $(\bar{c}_r, \bar{y}_r) \leftarrow$  mean cost and accuracy of rung  $r$  over  $T$   
2  $H \leftarrow$  upper convex hull of  $\{(\bar{c}_r, \bar{y}_r)\}_r$   
3 **reject** the pair if  $\bar{c}_R/\bar{c}_1 < 1.5$ : a hull over configurations  
that barely differ in cost is decided by a millisecond  
**gap**  
4  $(c_\pi, y_\pi) \leftarrow$  mean cost and accuracy of  $\pi$  over  $T$   
5  $\text{gap}(\pi) \leftarrow y_\pi - H(c_\pi)$   
**slack**, the part of that gap needing no signal  
6  $\hat{c}_r(x) \leftarrow$  *predicted* cost of rung  $r$  on  $x$ , never measured  
7  $\sigma \leftarrow$  greedily buy the cheapest rung upgrades under  $\hat{c}$   
until the budget  $c_\pi$  is spent  
8  $\text{slack} \leftarrow y_\sigma - H(c_\sigma)$   
**report**  
9  $\text{gap}(\pi)$  and slack, over resampled folds; the signal is credited  
with  $\text{gap}(\pi) - \text{slack}$  and no more

---

queries on a steeply priced model].

One reading of the split we cannot exclude: the how-far target separates on the two InfographicVQA pairs and on neither DocVQA nor ChartQA, which is also how a corpus-difficulty effect would present rather than a query-level signal. Separating the two needs a corpus whose pages vary in difficulty independently of their size [RESULT NEEDED].

*Gain dispersion* never exceeds 0.568 and we set it aside. *Answer length* reaches 0.600 on one pair, the crossed cell, which is also the pair cost allocation explains least; on a single cell we record that as a coincidence worth one experiment rather than a finding.

**The residual is therefore unexplained after a bounded search**, which is weaker than a mechanism and stronger than an admission. One further caveat: all four targets are functions of the recorded outcomes, so a mechanism operating through the serving stack rather than through the query, such as an interaction between page size and batching, would not appear in any of them. These AUROCs are reported with percentile intervals and are deliberately not entered into the family-wise correction of §12: their null is 0.5 rather than zero, and thirty-two tests asserting nothing would inflate the correction against the results we do assert.

**The floor this comparator relaxes.** Algorithm 2 states the whole procedure, including the rejection rule for degenerate pairs and the requirement that the signal-free policy price itself by predicted rather than measured cost. Proposition 2 takes the hull over fixed configurations priced at their mean cost, which is what this literature would compare against if it compared at all. The tight floor is the hull over policies measurable with respect to the per-example cost vector, of which ours is a relaxation. The measurements above bound the relaxation at +0.008 on seven of these eight pairs, so the two floors nearly coincide

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**Algorithm 3** Calibrated ladder escalation

---

**fit**, given rungs  $r = 1 \dots R$  in increasing cost  
1  $s_i \leftarrow$  any routing score read from the cheap pass  
2 **for** each rung  $r$ :  
3  $G_i^{(r)} \leftarrow \mathbf{1}[\neg y_i^{\text{cheap}} \wedge y_i^r] - \mathbf{1}[y_i^{\text{cheap}} \wedge \neg y_i^r]$   
4  $\hat{g}_r \leftarrow 2 \text{isotonic}(s, (G^{(r)} + 1)/2) - 1$   
**serve**  $x$ , given  $V$   
5  $\Delta t_r \leftarrow t_r(x) - t_{\text{cheap}}(x)$  for each rung  
6  $v_r \leftarrow V \hat{g}_r(s(x)) - \Delta t_r$   
7 **if**  $\max_r v_r > 0$  **then** run rung  $\arg \max_r v_r$   
8 **else** answer from the cheap pass

---

there; on the eighth, and on a serving stack whose price varies more sharply across queries they need not, and the cost-only column should be measured alongside any reported gap rather than assumed small.

## 10 How far to escalate, not just whether

Every escalation method we have read is binary: answer from a downsampled image, or run the full one. Taking the cheapest configuration that answers each query correctly says that collapses the decision carrying most of the money.

| cheapest rung that answers correctly | share        |
|--------------------------------------|--------------|
| 384 px (no escalation)               | 38.3%        |
| intermediate rung                    | <b>17.7%</b> |
| full resolution                      | 5.2%         |
| nothing we run                       | 38.8%        |

Of the queries escalation can repair, 77% are repaired by the intermediate rung. A binary policy over-serves three quarters of what it escalates. The rungs are also not equally efficient:  $384 \rightarrow 768$  buys 13.3 points for 79 ms (+1.69 points per second) where  $768 \rightarrow \text{full}$  buys 3.3 for 38 ms (+0.86). The binary jump measures +1.42, which is what averaging a good rung with a poor one gives.

**The rule generalises without modification.** Rung  $r$  has its own expected gain over answering cheap and its own extra latency, so

$$r^*(x) = \arg \max_r (V \mathbb{E}[G_r | x] - \Delta t_r), \quad (12)$$

answering cheap when no rung is positive. Eq. 12 reduces to Eq. 11 when there is one rung; the binary policy is the special case that has deleted the middle of its own ladder. One calibrator per rung is fitted on the same score, so the additional cost is a second isotonic fit over cached values.

Proposition 1 extends without change. Escalating to rung  $r$  shifts expected cost by  $\Delta J_r(x) = -w_e \mathbb{E}[G_r | x] + \lambda_t \Delta t_r$ , the choice is again pointwise because the objective is linear in the selection, and the best action at

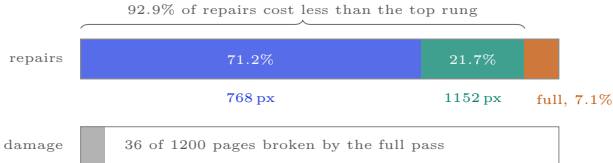


Figure 19: Where escalation pays on the ladder, over 1200 DocVQA pages. Of the 709 pages the cheap pass gets wrong and some rung gets right, 505 are already repaired one rung up and only 50 need the top rung, so a binary escalate-or-not decision buys most of its accuracy at several times the necessary price. The lower bar is what a repair count hides: 36 pages answered correctly at 384px are answered wrongly at full resolution, which is why the quantity to calibrate is a *signed* gain and why the escalated pass cannot be treated as attaining a fixed accuracy.

|         | top rung exists | accuracy | latency |
|---------|-----------------|----------|---------|
| DocVQA  | 98%             | +0.003   | −35.1   |
| V*Bench | 98%             | −0.008   | −3.5    |
| TextVQA | 0%              | −0.006   | +1.2    |
| VQAv2   | 0%              | −0.002   | −0.8    |

Table 21: Ladder minus binary, averaged over four operator preferences, latency in milliseconds. “Top rung exists” is the share of images on which the full-resolution pass spends more visual tokens than the intermediate one.

$x$  is the rung minimising  $\Delta J_r$ , or the cheap pass when every  $\Delta J_r$  is non-negative. Multiplying through by  $-1/\lambda_t$  gives Eq. 12. The binary rule is therefore not a simplification of the ladder but a *constrained* version of it, forced to choose between two of the rungs available to it.

**It pays where the ladder physically exists, and nowhere else.** Paired over four domains and four preferences, the ladder is  $-9.6$   $[-17.2, -2.6]$  ms cheaper at  $-0.003$   $[-0.008, +0.002]$  accuracy: significantly cheaper, indistinguishable in accuracy. The per-domain breakdown is the real result (Table 21).

On DocVQA the ladder saves 35 ms at no accuracy cost. On TextVQA and VQAv2 it does nothing, correctly, because for those datasets there is no top rung to find.

**A configuration name does not determine a configuration.** That last column is a measurement fact worth stating on its own. The processor caps its target at the input’s longest side, so on a small image the “full-resolution” pass *is* the intermediate pass: identical token count, identical cost, identical output. This holds for 100% of TextVQA and VQAv2 and 2% of the other two, 56% of the pilot. The escalation this paper studies is a real escalation for half its data.

This is the fourth error of the same family we have found, after processor upscaling, patch-grid quantisation,

| step       | net gain [95% CI]      | cost | points/s |
|------------|------------------------|------|----------|
| 64 → 320   | +0.383 [0.352, 0.414]  | 105  | +3.65    |
| 320 → 640  | +0.086 [0.060, 0.111]  | 129  | +0.67    |
| 640 → 1088 | −0.002 [−0.026, 0.022] | 156  | −0.01    |

Table 22: Rung-to-rung value on 1200 DocVQA pages, by visual-token count, latency in milliseconds. Accuracy peaks at 640 tokens (0.748) and 1088 tokens reaches 0.746: overlapping intervals for 70% more tokens.

and the flat escalation price of \$7. All four come from treating a configuration label as a cost or a capability. The general form of the lesson: in adaptive-resolution work, verify that the configurations differ *on the data*, not in the config file.

## 10.1 Where the ladder stops: a corpus ceiling

**A ladder built on measured rungs, and where it stops.** The mixture cannot price the second and third rungs, because for over half of it they are the same configuration. So we collected a single-domain pilot designed against that: 1200 DocVQA pages, with rungs chosen by measuring SmolVLM’s patch-grid buckets (64, 320, 640 and 1088 visual tokens) rather than by picking round pixel sizes. Every rung is strictly dearer than the one below it on 95 to 100% of images, and latency is affine in tokens at  $205.0 + 0.415v$  ms with a worst residual of 20.2 ms.

**What the taxonomy looks like on paper.** Figure 20 grounds the shares the ladder trades on, one real page per outcome. The thumbnail suffices on 27.9% of the 1200 pages, a higher rung flips a wrong answer to a right one on 49.9%, and 13.0% fail everywhere, full resolution included: the third panel fails by reading the wrong table row, an error more pixels cannot repair. The remaining 9.2% are non-monotone in the rung order, correct at 768px or at full resolution while 1152px fails.

The result is a saturation point, and it arrives early (Table 22).

The first rung is *five times* more efficient per millisecond than the second, and the second is the last one that pays: the top step repairs 8.1% of queries and damages 8.3%, a wash costing 156 ms. Of everything escalation can repair, 93% is repaired below the top rung and only 4.2% needs it. And this is measured on document pages, the regime where resolution matters most, so it is not a low-detail artefact.

**More pixels stop buying accuracy well below the resolution the model accepts.** That is a sharper form of the non-monotonicity in §3.3: escalation does not merely harm a few queries, it has a ceiling, and past it the harm rate catches the repair rate exactly.

A null result is only a result if it is tight, so it is worth saying that this one is. The interval on the top step has

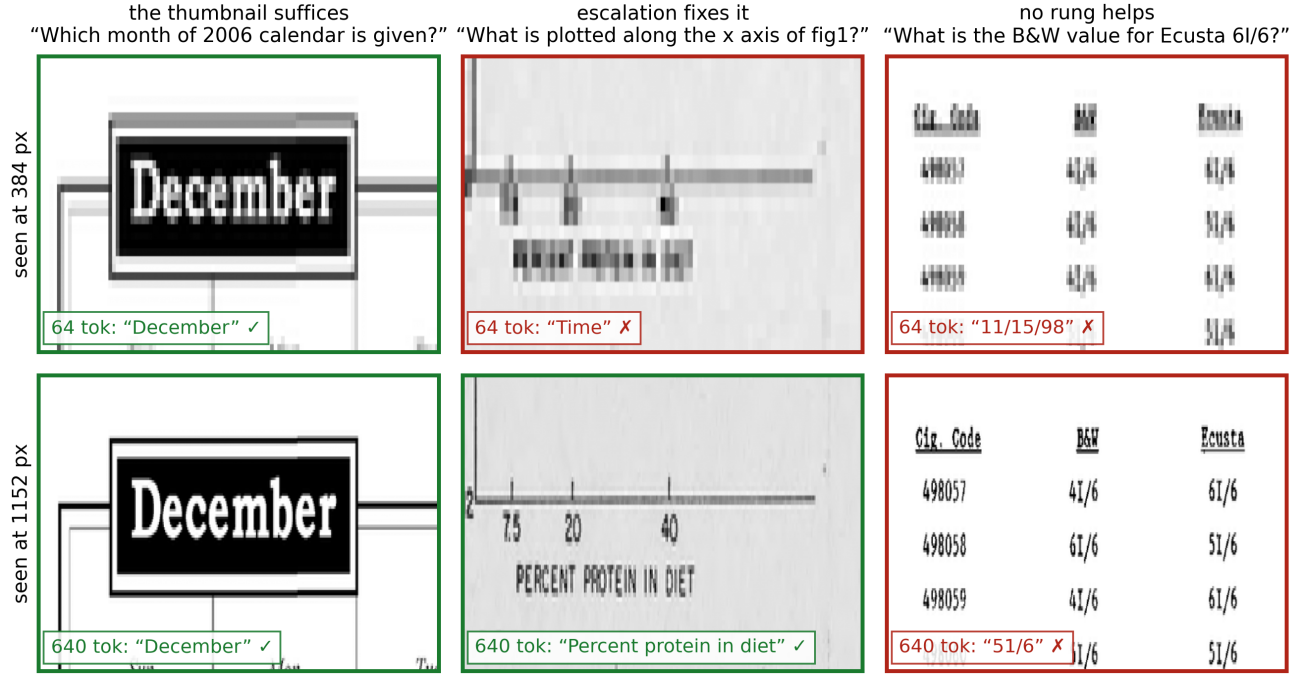


Figure 20: Three DocVQA pages rendered at the resolution each rung actually receives (Lanczos resize to the rung’s longest edge, shown nearest-neighbour; the model’s answer and its visual-token price are inset). Left: layout-scale evidence survives the thumbnail and 64 tokens already answer; this covers 27.9% of the 1200 pages. Centre: the axis label is an unreadable smudge at 384px and the model confabulates “Time”; at 1152px it reads. A higher rung repairs a wrong answer this way on 49.9% of pages. Right: the table resolves at 1152px yet the model reports the value of the wrong row, and full resolution repeats the error; 13.0% of pages fail at every rung, and for them the binding constraint is not legibility, so no resolution policy can help.

half-width 0.024, against measured gains of +0.383 and +0.086 on the steps below it: an effect a third the size of the smallest one we do detect would have been visible. That distinguishes this from a claim of no effect made at a sample size incapable of finding one, which is the failure mode we impose on ourselves elsewhere and check for here.

### The ceiling belongs to the corpus, not to the model.

A saturation point measured on one model could be that model running out of capacity, or the pages simply being legible at that resolution. The two explanations predict opposite things, so we re-ran the entire ladder on SmolVLM-256M over the same 1200 pages: capacity implies the smaller model saturates *earlier*, legibility implies the same rung with lower accuracy throughout (Table 23). A reviewer of an earlier draft made the obvious objection to the three models we had then, and it was right: they share a lineage and a data recipe, so a common pretraining corpus was an untested alternative cause. The fourth row answers it.

It is legibility. Halving the parameters costs accuracy everywhere and moves the ceiling nowhere, and all four top-step intervals are tight, so these are measured nulls rather than failures to measure. Plotted, the four curves

|                                                | 384   | 768   | 1152         | 2048  | top step |
|------------------------------------------------|-------|-------|--------------|-------|----------|
| SmolVLM-500M                                   | 0.279 | 0.662 | <b>0.748</b> | 0.746 | −0.002   |
| SmolVLM-256M                                   | 0.220 | 0.588 | <b>0.653</b> | 0.666 | +0.013   |
| SmolVLM2-2.2B                                  | 0.413 | 0.669 | <b>0.738</b> | 0.754 | +0.016   |
| Qwen2-VL-2B                                    | 0.600 | 0.850 | <b>0.890</b> | 0.892 | +0.002   |
| <i>median visual tokens spent at each rung</i> |       |       |              |       |          |
| SmolVLM-500M/256M                              | 64    | 320   | 640          | 1088  |          |
| SmolVLM2-2.2B                                  | 81    | 405   | <b>810</b>   | 1377  |          |
| Qwen2-VL-2B                                    | 154   | 567   | <b>1312</b>  | 4161  |          |

Table 23: ANLS accuracy by rung, four models on the same DocVQA pages (1200 for the SmolVLM family, the first 500 for Qwen2-VL-2B), columns labelled by the pixel target. Every top-step interval is a tight null:  $[-0.026, +0.022]$ ,  $[-0.012, +0.037]$ ,  $[-0.011, +0.041]$  and  $[-0.016, +0.020]$ , half-widths 0.018 to 0.026.

rise and then flatten at the same place (Fig. 21).

**The ceiling survives leaving the family.** Qwen2-VL-2B shares nothing with the SmolVLM line: a NaViT-style encoder that emits tokens in proportion to input pixels rather than quantising to a patch grid, a different language model, a different data recipe. It is also a far stronger reader of these pages, peaking at 0.892 against the family’s best 0.754. That matters for what the null can mean. A

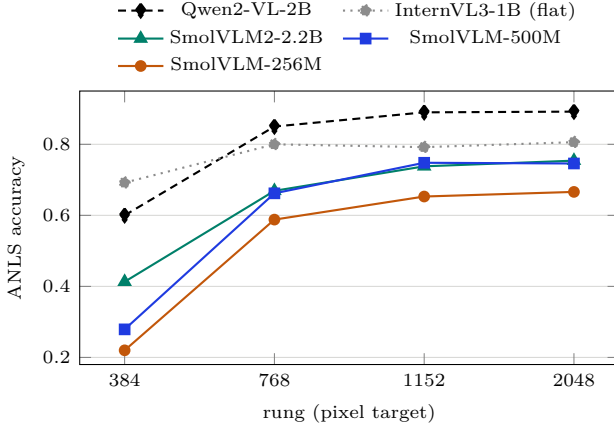


Figure 21: Accuracy against rung on the same DocVQA pages. The last step is flat for the four cost ladders, though they spend 640, 640, 810 and 1312 visual tokens respectively at the rung where it flattens, so no fixed token budget explains the ceiling, and the dashed curve is a model from another lineage entirely. The dotted curve is the control: InternVL3-1B spends the same tokens at every rung, and flattens one rung earlier, at 768 px. Its vertical position is not comparable to the others, since its cheapest rung already spends a full budget on an upscaled image.

model that stops gaining because it is too weak to exploit more pixels would prove nothing; this one stops gaining while answering 14 points more of the corpus correctly than any model in the family, and while being handed  $3.2\times$  the visual tokens to do it with (1312 at the ceiling, 4161 at full resolution) for a measured  $+0.002 [-0.016, +0.020]$ . Whatever is left unanswered at 1152px on these pages is not waiting on resolution for any of the four models, which points at the documents rather than at the models reading them.

## 10.2 Which axis binds, pixels or tokens

**The ceiling is not a token budget.** The two SmolVLM models share an 86M vision encoder, so “stops gaining at 1152px” and “stops gaining at 640 visual tokens” name the same point for both and cannot be told apart. SmolVLM2-2.2B tokenises differently, spending 810 tokens where the others spend 640, which separates them, and Qwen2-VL-2B spends 1312 there, a  $2.05\times$  spread across the four. All four stop at the same *pixel* target while spending very different numbers of tokens to reach it, which no single sequence-length ceiling explains.

**But it is not purely a pixel count either, and one model says so.** The obvious next sentence, that what runs out is the information in the pixels, is the one we wrote in an earlier draft and cannot support. In all four models the two quantities travel together: raising the pixel

| step        | net gain [95% CI]       | tokens | verdict |
|-------------|-------------------------|--------|---------|
| 384 → 768   | +0.108 [+0.070, +0.148] | +16    | gain    |
| 768 → 1152  | −0.008 [−0.026, +0.010] | +20    | null    |
| 1152 → full | +0.014 [−0.002, +0.030] | +38    | null    |

Table 24: InternVL3-1B on 500 DocVQA pages, where the token budget is near constant and only the input pixels move. Accuracy is 0.692, 0.800, 0.792, 0.806 by rung. Both nulls are tight, half-widths 0.018 and 0.016.

target raises the token count, so every rung changes both, and the argument above rules out a *fixed* token budget without establishing that tokens do no work. Separating them needs a model that spends the same tokens on a small image as on a large one, and InternVL3-1B is one by accident of its design. It selects a patch grid from the aspect ratio rather than the resolution, so a page downsampled to 384px is upscaled again to fill the same tiles as the 2048px original. Across 500 pages all four rungs are token-identical on 95% of them and the mean spend moves by  $1.02\times$  against  $5.3\times$  in input pixels: near enough to constant to read the ladder as pixels alone (Table 24).

With the budget held still, the gain stops one rung earlier, at 768px. The  $768 \rightarrow 1152$  step, worth  $+0.086$  to SmolVLM-500M and  $+0.040$  to Qwen2-VL-2B, is worth  $-0.008 [-0.026, +0.010]$  here. Two readings are open and one model cannot close them: either the pixel information of these pages is exhausted by 768px and what the other four bought in that step was their growing token budget, or InternVL simply reads less of the page. The second is the weaker of the two, because InternVL is not the weaker model. It peaks at 0.806 against SmolVLM-500M’s 0.748, so a model that reads *more* of this corpus overall is the one that stops gaining from pixels first.

**The other axis, with the pixels held still.** The control above moves pixels and holds tokens; the pair is only complete with its converse. InternVL selects its tile grid under a bound, so raising the bound multiplies the visual tokens spent on an image whose pixels never change. Pinning the input at full resolution and sweeping the bound over 500 pages measures what sequence length alone buys here (Table 25).

Tripling the token budget at fixed pixels buys  $+0.024 [-0.006, +0.054]$ , a null tight enough to have found an effect a third the size of the pixel gains we do detect. Spending a further 2258 tokens is associated with 0.046 *less* accuracy where it is actually spent; that sign is nominally significant and does not survive the family-wise correction of §12 ( $p = 0.023$ , adjusted 0.372), so we report it as an observation consistent with the premise of the token-pruning literature [4] and with a mechanism of interpolation rather than information, and not as an established effect. The pages where the bound changed nothing form an internal control that behaves exactly as

| step                                               | pages | extra tokens | net gain [95% CI]       |
|----------------------------------------------------|-------|--------------|-------------------------|
| 1280 → 3328 tokens                                 | 500   | +2011        | +0.024 [−0.006, +0.054] |
| <i>next bound, split by what it actually spent</i> |       |              |                         |
| tokens rose                                        | 240   | +2258        | −0.046 [−0.087, −0.008] |
| tokens unchanged                                   | 260   | 0            | +0.000 [+0.000, +0.000] |

Table 25: InternVL3-1B on 500 DocVQA pages with the input pinned at full resolution and only the tile bound moving. Accuracy is 0.782, 0.806, 0.784 by bound. The second bound raises the token count on 48% of pages and leaves the rest untouched, so the step is reported conditioned on what was spent rather than on the bound’s name.

greedy decoding requires, +0.000 [+0.000, +0.000] over 260 pages, which bounds the measurement noise in the other subset.

**Read against the fixed-budget ladder, this separates the two quantities.** Holding tokens still and raising pixels from 384 to 768 buys +0.108, which survives correction; holding pixels still and tripling tokens buys +0.024 with an interval spanning zero, and spending beyond that does not help. On this corpus the binding constraint is the information in the pixels, and sequence length is not scarce. That is the sentence an earlier draft asserted without evidence, then withdrew for want of it, and it is now supported by a pair of controls rather than by the co-varying ladders that cannot speak to it.

**What we are entitled to say** is therefore narrower than what we first wrote, and better supported. No single token budget explains the ceiling; pixel information alone is spent by 768px on the model that lets us isolate it; and additional sequence length, isolated in turn, does not repay its cost on this corpus. What we cannot do is attribute the 768 to 1152px gains of the four co-varying models between the two axes, since both move at every rung there. The serving consequence is untouched either way, because it never depended on the mechanism: every model we tested stops paying by 1152px, and Algorithm 4 finds that point without needing to know why it is there.

**That makes the saturation point reusable, as a bound.** Across these four it survives a change of serving model, of vision encoder and of lineage, which is already a more useful object than a threshold re-derived per model. It is not a constant, and the fifth model is why: with the token budget held flat the ceiling sits a rung lower, at 768px. The direction is the part worth keeping. No model we tested needs *more* resolution than the corpus ceiling, and one needs less, so the ceiling measured this way is an upper bound on what any model needs and running Algorithm 4 against the model actually being served can only move it down. A serving stack that caps escalation at the corpus ceiling is never wrong about the top of its ladder, only sometimes generous. We note in fairness that three of the four put a positive point estimate on the top step (+0.013, +0.016 and +0.002), leaning the opposite way from the 500M’s −0.002; at these interval widths that

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#### Algorithm 4 Locating a corpus’s resolution ceiling

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- 0 check the serving model is not the binding constraint on this corpus: a null at the top step is a ceiling only if a model reaching materially higher accuracy agrees with it
  - 1 draw  $n$  pages from the corpus ( $n \geq 300$ , Table 26)
  - 2 for each candidate pixel target  $p$ , *measure* the visual tokens the serving model spends on each page at  $p$
  - 3 keep the targets whose token counts are strictly increasing on  $\geq 95\%$  of pages; discard the rest as duplicates
  - 4 run every kept rung; record correctness and latency
  - 5 for each adjacent pair  $(r, r')$ , bootstrap the paired gain  $\bar{G}_{r \rightarrow r'}$  and its 95% interval
  - 6 **if** any interval has half-width  $> 0.05$  **then**  
report undetermined and collect more pages
  - 7 ceiling  $\leftarrow$  highest  $r'$  whose interval excludes 0
  - 8 serve with every rung above the ceiling removed  
(run on the served model: the corpus ceiling bounds it above)
- 

is noise, and we report it rather than claim a symmetry the data does not show.

### 10.3 Locating the ceiling, and what that costs

**The procedure, and what it costs to run.** If the ceiling is the transferable object then it has to be findable on a corpus nobody has measured, so we state the procedure we followed rather than only its output (Algorithm 4). Two of its steps encode failures this paper documents elsewhere. Step 2 measures the visual tokens each pixel target actually produces instead of trusting the target, because four measurement errors here arose from a configuration name that did not determine a cost, and on our own pilot two nominally different rungs were the same configuration for 56% of examples. Step 6 refuses to declare a ceiling from a wide interval, the failure mode that produced a premature verdict at  $n = 244$  on an earlier pass and again at  $n \approx 100$  mid-collection on the out-of-lineage run.

The cost of running it is small and measurable. Sub-sampling the pilot and recomputing the top-step interval gives the precision a given sample buys (Table 26): the half-width falls as  $0.78/\sqrt{n}$ , so the 0.05 bar of step 6 is reached at about 244 pages, and 300 pages name the same ceiling rung as the full 1200 on 93% of draws. The row that matters most is the first. At  $n = 100$  the procedure agrees with the full corpus barely half the time, which is not a ceiling measurement but a coin flip, and it is exactly the trap a practitioner would walk into by eyeballing a small pilot.

**The procedure on a corpus it has not seen, and what a second model does to the answer.** An earlier version of this paper predicted that a benchmark of denser type should place the ceiling higher than DocVQA’s 1152px. We collected one and ran Algorithm 4 on it unchanged: 500 InfographicVQA pages, the same task, the

| pages | half-width | p90   | names the same ceiling |
|-------|------------|-------|------------------------|
| 100   | 0.080      | 0.090 | 47%                    |
| 200   | 0.055      | 0.062 | 80%                    |
| 300   | 0.045      | 0.050 | 93%                    |
| 500   | 0.035      | 0.038 | 100%                   |
| 800   | 0.028      | 0.029 | 100%                   |
| 1200  | 0.023      | 0.024 | 100%                   |

Table 26: What a sample size buys when locating the ceiling, 200 draws per row on the 1200-page pilot. Half-width is of the paired top-step interval; the last column is how often a run at that size names the rung the full corpus names.

| corpus, model                                    | 384   | 768   | 1152         | 2048         | top step      |
|--------------------------------------------------|-------|-------|--------------|--------------|---------------|
| DocVQA, 500M                                     | 0.279 | 0.662 | <b>0.748</b> | 0.746        | -0.002        |
| InfoVQA, 500M                                    | 0.230 | 0.282 | <b>0.368</b> | 0.360        | -0.008        |
| InfoVQA, Qwen2-VL-2B                             | 0.350 | 0.488 | 0.580        | <b>0.650</b> | <b>+0.070</b> |
| <i>median visual tokens spent at each target</i> |       |       |              |              |               |
| DocVQA, 500M                                     | 64    | 320   | 640          | 1088         |               |
| InfoVQA, 500M                                    | 64    | 192   | 320          | 576          |               |
| InfoVQA, Qwen2-VL-2B                             | 70    | 297   | 609          | 1312         |               |

Table 27: The same procedure on two corpora and, on the second, two models. The 500M top steps are tight nulls  $[-0.024, +0.022]$  and  $[-0.044, +0.028]$ ; the stronger model’s is a gain of  $+0.070 [+0.032, +0.110]$ , so on Info-graphicVQA the ceiling lies above the highest rung we run.

same ANLS metric, the same pixel targets. Read with the 500M model the ceiling did not move. Read with a second and much stronger model it moves past the top rung we run (Table 27), and that difference is the section’s result.

**A null at the top step is only a corpus ceiling if the model is not the binding constraint, and this is the counterexample.** On Info-graphicVQA the 500M model peaks at 0.368 and stops gaining at 1152 px. Qwen2-VL-2B reads the same pages to 0.650 and is still gaining  $+0.070$  at the top rung. The weaker model’s null was its own capacity, not the pages’ legibility, and Algorithm 4 as stated would have called it a corpus ceiling. **We therefore withdraw the claim, made in an earlier version of this section, that two corpora share a 1152 px ceiling.** What holds is narrower: DocVQA has a ceiling at 1152 px that four models of very different strength agree on, including one that reads it to 0.892 and is therefore not capacity-limited there; Info-graphicVQA has no ceiling we have located, and the rungs we run do not reach it.

**The failure hands us the missing precondition.** Algorithm 4 locates where a *model-corpus pair* stops paying, and that is a corpus property only when the model is strong enough on the corpus for legibility to bind before capacity does. We add that as step 0: before reading a null as a ceiling, check that at least one model reaching a substantially higher accuracy agrees with it, or report the ceiling as specific to the pair. The DocVQA ceiling

| V    | accuracy delta          | latency delta (ms)             |
|------|-------------------------|--------------------------------|
| 400  | +0.359 [+0.346, +0.374] | +86.0 [+84.3, +87.4]           |
| 800  | -0.010 [-0.019, -0.001] | -161.1 [-163.9, -158.4]        |
| 1600 | -0.010 [-0.017, -0.003] | -154.5 [-158.3, -150.7]        |
| 3200 | -0.002 [-0.009, +0.005] | <b>-136.4 [-141.1, -132.0]</b> |

Table 28: Ladder minus binary on 1200 DocVQA pages, paired within each operating point over 30 resamples. At  $V = 400$  the binary rule barely escalates at all and the ladder buys 36 accuracy points for 86 ms; above that it trades at most one accuracy point for roughly 150 ms.

passes this test, since the 500M model at 0.748 and Qwen at 0.892 agree on it. The Info-graphicVQA measurement fails it, and would have been published as a corpus ceiling had we not run the second model. The cost of the check is one additional model on the corpus, which is what the rest of this paper spends on every claim it makes twice.

**The ladder rule finds the ceiling on its own.** Given rungs and prices, Eq. 12 selects the top rung on 0 to 6% of queries across the whole preference range: it discovers the saturation from the calibrated gains without being told. Paired against the binary policy over 30 resamples of a held-out 300 (Table 28), it is significantly cheaper at three of four operating points, and at the highest preference it is cheaper at accuracy the comparison cannot separate.

The mixture measured this effect at  $-9.6$  ms because most of its ladder did not exist. We report the intervals per operating point rather than pooled across them: an earlier version of this comparison bootstrapped the four preference means, which resamples four numbers and yields an interval four points wide.

**A refinement of ours that does not pay.** The cost audit suggested an obvious improvement: since escalation prices vary  $2.5\times$  across the pilot, charge each query its own break-even  $\tau_i = \Delta t_i / V$  rather than a global one. Implemented and measured, it is worth nothing,  $+0.00002 [-0.0003, +0.0004]$  accuracy and  $-1.5 [-4.3, +0.4]$  ms, both spanning zero.

The reason is the confound again. Image size is largely a *domain* property, so within a domain the prices a per-query rule would discriminate between barely differ, and there is nothing for the refinement to bite on. It would matter for a router serving one queue of genuinely mixed image sizes, and we report it as a negative rather than dropping it because the negative is the informative part: the same fact that makes the probe a domain detector makes per-query pricing redundant within a domain.

**One caveat on the UCCI contrast.** The gap of Table 17, calibrating correctness over-spending against calibrating the gain, is a mixture effect too. Within a domain both rules land on the front at different operating points, and the sign flip that separates them is weaker. We state

the contrast as measured on the mixture and unresolved within a domain.

## 11 Discussion

**What an evaluation has to include before a signal can be believed.** The measurements above converge on three requirements, each of which changed a conclusion here when applied. Score signals *within* each distribution rather than across a mixture: pooling inflated our probe from a within-domain 0.572 to a pooled 0.760, and the inflation is the mixture’s own heterogeneity. Charge every signal what it costs to read: entropy ranks respectably and still fails the no-signal baseline, because reading it requires finishing the pass it is meant to price. And compare against randomisation between fixed configurations, not against a single fixed endpoint, since the endpoints are the two least interesting points of a reachable segment. None of the three is expensive. Together they are the difference between the result we first believed and the one we can defend.

**What this means for the systems we compare against.** It bounds rather than refutes them. The trained policies of [1, 2, 3] operate at 7B, where the untrained self-report we can measure is worth little (Table 6) and where the training is presumably what closes that gap. Our finding is about what is available *without* training below one billion parameters, and there the honest answer on mixed traffic is that the file header is competitive with the residual stream. That is a statement about the difficulty of the sub-1B regime, not about the value of learning the decision. It does suggest one cheap control for that literature: report the escalation policy’s accuracy against a policy that knows only the benchmark identity, which on our mixture accounts for the whole of the adaptive gain.

**Where the remaining money is.** Not in deciding *whether* to escalate on heterogeneous traffic, where a free descriptor recovers most of the available gain, and not in a third abstention action, which the calibrated rule already suppresses (§E). It is in deciding *how far*, which is worth 35 ms per query at no accuracy cost on document traffic, and in knowing where to stop, which is a property of the corpus rather than of the query and needs measuring once. A per-query signal earns its keep exactly where domain identity carries no information, and there it is output entropy rather than the probe that survives. The practical rule is therefore a two-line one: measure how much of the escalation value your traffic explains with free metadata; if most of it, route on the metadata and spend the engineering on the ladder; if little, read the model and give it graded actions.

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### Algorithm 5 Choosing a routing architecture for a workload

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**given** a sample of the traffic with its existing labels,  
 and two configurations whose cost differs by at least  $1.5\times$   
**measure, before building anything**  
 1 run the cheap and the expensive configuration on the sample  
 2  $G_i \leftarrow \mathbf{1}[\neg y_i^{\text{cheap}} \wedge y_i^{\text{full}}] - \mathbf{1}[y_i^{\text{cheap}} \wedge \neg y_i^{\text{full}}]$   
 3  $a \leftarrow \text{AUROC}(\text{longest image side}, \mathbf{1}[G_i > 0])$   
**read it as an ordering, not as a cut**  
 4 **if**  $a$  is far above chance (0.748 on our four-dataset mixture): escalation value is carried by *which kind of input* arrives. Route on the free metadata and spend the engineering on the ladder  
 5 **if**  $a$  is near chance (0.508 to 0.533 on our three single workloads): that axis is absent. A useful signal must be read from the model, with graded actions  
**then, on either branch**  
 6 locate the corpus ceiling once (Algorithm 4), and price the policy against the no-signal floor (Algorithm 2)  
**caveat**  
 7 four points do not calibrate a threshold, and traffic of intermediate heterogeneity is what we have not collected

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**A diagnostic, offered as an ordering test.** The practical question a reader faces is which regime their traffic is in, and our measurements supply a candidate test that costs nothing: score raw image size against the escalation gain on a sample of the traffic, using the labels the sample already carries. It reaches 0.748 on our heterogeneous mixture and 0.508, 0.508 and 0.533 on the three single workloads, so a free descriptor that ranks near chance is a sign that the between-domain axis is absent and that whatever value remains has to be read from the model. Four points do not establish a threshold, and traffic of intermediate heterogeneity is exactly what we have not collected, so this is an ordering test rather than a calibrated cut, and calibrating it would need mixtures sampled between our two extremes. It is cheap enough to run before committing to an architecture, which is the property that makes it worth stating at all. Algorithm 5 states it together with the decision it feeds.

**Status of each claim.** Table 29 separates what the evidence supports at three different strengths, since the paper’s claims are not all of one kind.

## 12 Limitations

Five serving models, one GPU, and two pilots: 1000 examples across four datasets and 1200 within one, the latter run on all five. The saturation ceiling is the claim we have pushed hardest, over an  $8.6\times$  parameter range, three vision-encoder mechanisms and two lineages. It held across four of the five and moved down a rung on the fifth, the one whose token budget does not follow its input resolution, which is why we state it as a bound rather than a constant. It is established on one corpus, DocVQA,

where four models spanning  $8.6\times$  parameters agree on 1152px and the strongest of them reads the corpus to 0.892, so legibility rather than capacity is what binds. On our second corpus it is not established: the 500M model’s null there is its own capacity, and a stronger model keeps gaining past our highest rung, so InfographicVQA’s ceiling lies above 2048px and we have not located it. That failure is reported in §10 because it supplies the precondition Algorithm 4 lacked, and because we would have published the weak model’s null as a corpus property had we not run the second. What we claim transfers is the procedure, now including that precondition, rather than the number. The mechanism behind the ceiling is settled only on the model that lets both axes be moved separately. There, pixels at fixed tokens buy +0.108 over one rung and tokens at fixed pixels buy +0.024 with an interval spanning zero. We do not claim the same decomposition for the four models whose tokeniser follows resolution, because every rung of theirs moves both quantities, and a corpus with finer print could shift the balance. Our strongest baseline is the published escalation *mechanism* without its training; a re-implemented VisionThink at sub-1B scale remains the comparison the field would want, and the gap between an untrained self-report and a trained one is exactly what we cannot measure.

Five limitations deserve more than a mention. Every comparison in this paper is a cost comparison and the single-domain pilots time each pass once, so the per-example escalation price our policies read carries measurement error we did not originally bound. That is not hypothetical: §9 reports a policy that harvested exactly this noise and the retraction it forced. We re-time a subsample to bound it, and the bound refuted the account we first gave for it: the measurement error is a fifth of the variation our policies read, and the anomalies we had blamed on it reproduce. The latency saving needs an inference stack that can abandon a prefill mid-flight and start a different image. We verified this is real rather than arithmetic: physically truncating the decoder to its first six layers gives a 20.5ms prefill, within 1% of Eq. 7’s prediction, and an escalated query then costs 277.0ms against 402.9ms for completing the cheap pass first, a measured 31% saving. What standard serving frameworks do not expose is the *control* to do this per query, so the saving is available but not yet convenient. Absolute accuracies run from 0.385 to 0.580, a regime where the model is wrong about half the time, and conclusions drawn there may not transfer to a model that is usually right. The probe’s direction does not transfer across question domains (trained on detail-limited data and tested on knowledge-limited data it scores 0.422, and the reverse 0.366, both below chance) so deployment needs a domain-representative calibration set and a shift detector. And the cross-model transfer result uses two models that share an identical 86M vision encoder, so part of what transfers may be encoder-side rather than genuinely cross-model; separating those requires a source model with a different encoder.

We ask many questions of one dataset mixture, so the intervals above are corrected rather than nominal. Two hundred and twenty-seven paired comparisons at the nominal 5% level make at least one false positive close to certain, so we convert each bootstrap to a two-sided p-value and run Holm’s step-down over the whole family. Every comparison carries its own resample vector, so all of them are scored by the same bootstrap rather than by an approximation to a reported interval.

One hundred and sixty-three of the two hundred and twenty-seven clear the nominal level and one hundred and thirty-nine survive the step-down. Twenty-four claims are lost, and they are worth naming individually because six of them are convenient for our thesis. Five of the twenty-four are loopholes we measured against ourselves: the graded cost-only slack adjusts out on four of the eight pairs and the binary one on a fifth, so on those pairs the comparator has no measurable gap at all and the clearances there stand on firmer ground than the correction elsewhere costs us. That is worth stating plainly, because the correction is usually reported as what a paper loses. The probe’s AUROC advantage on the recovery target adjusts from  $p = 0.041$  to 0.662. Two preferences at which the probe nominally beats the free image descriptor adjust to 0.612 and 0.426, which makes the two signals harder to distinguish rather than easier. Two of the points at which the free descriptor sits *below* the hull inside one workload adjust to 0.130 and 0.372, which weakens that row to indistinguishable from the hull rather than worse than it: the correct reading is that the descriptor buys nothing there, not that it is harmful. Both readings of the harmful top step in the tile-budget control adjust to 0.745 and 0.722. The second corpus’s weakest rung gain, +0.052 from 384 to 768px on InfographicVQA, adjusts to 0.331; the step that sets that corpus’s ceiling, +0.086 from 768 to 1152px, survives, so the ceiling determination does not depend on the lost one. And the single nominal clearance of the hull on InfographicVQA adjusts from 0.004 to 0.108, which is why §10 states the positive half of the inversion on two workloads rather than three. Every cost comparison survives, as do the pixel-axis gains, the domain-identity bound, the free descriptor’s clearance of the hull at every operating point of both costings, and the entropy ladder’s clearance inside the DocVQA workload. We keep the eight in the paper as observations and no longer state them as established.

The split is itself informative. **Every surviving comparison is about cost**; all three probe-versus-entropy *accuracy* comparisons fail, as does the top rung’s gain. The split matches the paper’s claim structure: what we demonstrate is cheaper compute at accuracy the data cannot separate, not better answers.

The Pareto-dominance result is the one we have tested hardest and it came back qualified: universal dominance holds on the reported fold and in 52% of resplits, so the claim we make is the distributional one (90% of the front, a saving positive in 200 of 200 splits) and not the categor-

ical one. Readers should treat any single-fold dominance statement in this literature, including our own earlier one, as untested until resampled.

One limit is specific to §8:  $V$  is elicited, not measured, so the contribution is that the policy is determined *given* a preference, not that we know the preference. The adversarial result is a proxy: it shows the probe is not incidentally moved by what moves entropy, not that it resists an attack optimised against it.

The confound audit of §7 has a limit of its own, in our favour and worth stating. Fitting inside one dataset gives the probe 210 training examples where the pooled fit gets 600, so part of the collapse could be sample size rather than absent signal. Two observations argue against that reading (entropy needs no training at all and is therefore unaffected, and the probe is at or below chance rather than merely noisy) but the clean test is a single-domain corpus large enough to fit and test inside, and 300 examples per dataset is not it. We report the confound as established and its magnitude as uncertain.

## 13 Reproducibility

Every numeric claim in this paper is re-derived from the run records by a validation harness with an explicit threshold per claim, which exits non-zero if any stops holding; it currently checks eighty, including the negative results on localization, on our own energy instrument, and on the universality of Pareto dominance, that last one asserts the claim is *not* universal, so restating it too strongly fails the build. Claims stated in the text and not covered by a check are marked as unverified. The oracle runs are stored as append-only JSONL with per-configuration answers, generation scores and hardware measurements, so correctness metrics and cost weights can be changed offline without re-running any model. Splits are hash-based on example identifiers, so growing the pilot never reshuffles examples already assigned. Decoding is greedy throughout (16 new tokens, no sampling), so every accuracy in this paper is deterministic given the checkpoint and the image. The single-domain pilots time each pass once; the cost model is fitted on per-bucket medians over at least 50 passes, so no individual timing can set the line. The discovery that 56% of the mixture’s full-resolution passes were not distinct configurations (§10) is a protocol defect of ours that the verified ladder of the single-domain pilot was designed to remove.

**Compute.** Every experiment in this paper runs on one RTX 4060 laptop GPU with 8.6 GB of memory. Whether the component shares of §3.2 hold on the embedded hardware this line of work targets is something one GPU cannot settle; Amanzhol and Park [22] measure vision-transformer inference on both a Jetson TX2 and a mobile GPU and report rankings that differ between the two devices, which is the reason we state our cost model as measured on our

hardware rather than as a property of the architectures. The oracle runs comprise 54 660 instrumented generations totalling 305 minutes of measured model time, of which the 1200-page single-domain pilot is 6 000 generations per serving model, 40 minutes each for the two SmolVLM models and 54 for SmolVLM2-2.2B; activation extraction and the component profiles add roughly a further half hour. Wall-clock is appreciably larger, since image decoding, the processor, and the instrumentation itself sit outside the timed region. No training of any kind was performed: every probe here is a difference of means over cached activations, and every calibration a single isotonic pass, both of which take seconds on a CPU. The reproduction cost of this paper is therefore about six GPU-hours, which is the point of working at this scale.

## 14 Conclusion

We set out to ask what a 500M vision-language model knows about its own need for more pixels before it generates anything. Measured against baselines that need no model at all, the defensible answer is narrow, and the baselines are the reason it is worth stating.

On a four-dataset mixture, a linear probe on pre-generation activations ranks escalation value better than output confidence and yields a policy readable after a sixth of the language model. Inside any one of those datasets it reaches 0.572 where output confidence, fitted on nothing, reaches 0.663, and more training data narrows that gap without closing it. What the direction mostly encodes is which *kind* of image is present: a quantity that predicts escalation value across a benchmark mixture, because the mixture’s datasets differ enormously in how much resolution buys, and predicts much less within one. Raw image size, free to read, carries much of the same information: built into a policy and priced, it clears the randomisation hull at every operating point of either costing on the 500M model and stays within 0.010 of the probe there, though on a 256M model the probe keeps an advantage at tight budgets. A policy handed the dataset label and escalating at random within each dataset dominates both, which places an upper bound on the whole exercise: on mixed traffic, the adaptive gain available is the gain from knowing which benchmark is being served.

Three things bound the claim rather than one. This is a way of deciding *whether* to look again and not *where*; on the evidence here it is a way of deciding for a heterogeneous mixture and not for a workload; and on a mixture it is not a way that needs the model’s activations at all. Inside a workload the ranking inverts: the free descriptor becomes the weaker value ranker on all eight corpus-model pairs, and a model-read signal on a graded action space clears the no-signal baseline on five of them. The descriptor’s policy still clears where the serving model prices resolution steeply, about half of it by allocating budget rather than by ranking anything, which is a property of the comparator

we measured rather than a capability of the descriptor. Neither regime is the general case, and that is the paper’s first result.

What survives is the part that did not depend on the probe. The decision rule is signal-agnostic and survives on output confidence, where one stated preference produces a different escalation rate per domain without retuning. The resolution ladder carries most of the money that was attributed to the escalate/answer decision: on document pages, 93% of what escalation repairs is repaired below the top rung, and the ceiling sits at a pixel resolution that four models with three tokenisers and two lineages agree on. A fifth, whose token budget does not follow its input resolution, stops one rung earlier, so that ceiling is the most any model needs rather than what each one needs.

The comparison that decides whether any of this matters is against randomising between fixed configurations, which needs no signal. The standard entropy threshold never clears that bar on our mixture; the probe clears it only because it is read mid-prefill; the ladder clears it on homogeneous traffic, by margins that survive the largest gap a policy reading only cost can reach. Where a single fixed rung fits the budget, pinning it at the corpus’s measured saturation point is the right policy, and per-query routing is for the traffic no single rung fits. Both need the ceiling measured rather than guessed, which is why we state the procedure and its price alongside the number.

The result we would most like others to carry away is the negative one in §5.1, which we encountered four times. A probe trained on the conditional target beat its baseline on ranking and lost the policy comparison by 8.5 points. An ensemble of entropy and probe then beat both on ranking and lost the policy comparison on cost. A rule calibrating the weak pass’s error probability, which is the target [11] calibrates, matched our accuracy within two points while spending 59% relatively more compute. And the signal that beat all of them on the mixture turned out to be ranking the mixture. The target was wrong, then the signal was too expensive to read, then the calibration target was wrong, then the evaluation distribution was doing the work. None of the four is visible in an AUROC table.

Reporting ranking quality without building the policy is reporting the wrong number. The four ways of getting it wrong are optimising a target the policy does not face, optimising a signal the policy cannot afford, calibrating a quantity whose assumptions the setting violates, and reading a between-group difference as a within-group one. All are easy to miss.

The last generalises furthest, with the caveat that the literature we borrowed the probe from had already avoided it: any probe evaluated on a mixture of benchmarks that differ in the outcome being predicted should be scored within each of them before it is believed. The text-routing work reports per-dataset AUROC and warns explicitly about directions that capture dataset-specific cues [6]; we pooled, and pooling is what produced our headline.

The check costs one line, and it was our own evaluation protocol rather than the method that failed it.

## A Knowing *whether* is not knowing *where*

Our measurements show region choice matters more than action choice, so we tested whether the same internal signals locate the region. SmolVLM lays its 64 visual tokens in a contiguous  $8 \times 8$  grid, so the states covering candidate cell  $k$  can be pooled,

$$z_k = \frac{1}{|\mathcal{P}_k|} \sum_{p \in \mathcal{P}_k} h_p^{(\ell)}, \quad (13)$$

and ranked by a probe trained on which cells actually answered correctly. This requires no judge model, no reinforcement learning and no extra forward pass.

On a  $2 \times 2$  grid the localizer scores 48 to 51% across six depths against 51% for choosing uniformly at random, with a 64% oracle ceiling. Overlapping cells make random choice strong there, so we repeated the experiment at  $3 \times 3$  and then at  $4 \times 4$ . The finer grid is a fairer test: on the 400 examples all three share, the gap between random and the oracle widens from 12.2 to 16.2 points as cells shrink, so a localizer has more to find, not less.

**One split is not enough to state a negative.** Our earlier claim, that at no depth on either grid does the localizer beat random, rests on a single train/test split. Resampling it 60 times changes the sign of the answer (Table 30).

At  $4 \times 4$  the localizer beats random at every depth, with intervals excluding zero. It also beats it by 0.9 accuracy points, which is 6% of the gap that a perfect localizer would close. **The honest claim is about magnitude, not sign:** the signal is real and it is too small to build on. We state it that way because we have insisted elsewhere in this paper that a null is only a result when the interval is tight enough to have found an effect, and the converse obligation applies here.

**What the localizer would have been worth.** A negative result is easier to accept when the thing that failed is cheap. This one is not. Both ways of spending a visual-token budget are in the pilot, so they can be put on one axis (Table 31): a rung spends its tokens on the whole frame, a crop spends them on one cell at high magnification. This is also the comparison §3.2 has been making against the pruning literature [4] in words rather than numbers.

The two readings of the crop row carry the comparison (Fig. 22). A *randomly* chosen cell reaches 0.392, which is what the same tokens buy in resolution (0.383 at 64, and about 0.42 interpolated at 128). The *best* cell reaches 0.584, beating the dearest resolution configuration while

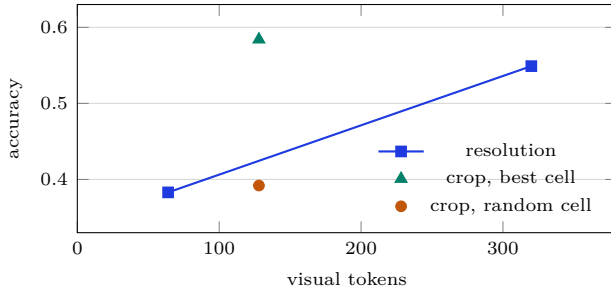


Figure 22: Two ways to spend a visual-token budget,  $n = 1000$ . The best crop sits above the resolution line at  $2.5\times$  fewer tokens; a randomly chosen one sits on it. The vertical distance between the two crop markers is what a localizer would have to recover, and §A recovers 6% of it.

spending  $2.5\times$  fewer tokens. **Position is worth more per token than resolution, and only to a policy that knows where to look.**

That prices the failure. A working localizer would be the most token-efficient action in this action space, which is presumably why [3] is willing to spend cold-start fine-tuning, multi-turn GRPO and a 70B judge to obtain one. Our 0.9-point margin buys none of that headroom.

The low-resolution pass knows *whether* more resolution would help and essentially not *where* to spend it. This bounds what internal signals deliver at this scale, and suggests that the supervision machinery of [3] is not overengineering.

## B What each part is worth

The components above are each justified where they are introduced, over five sections and on folds and cost models that were not always identical. That is how a paper acquires an inconsistency it cannot see, so we recompute all of them against one reference on one fold with one cost model.

**Read at a fixed preference, an ablation says the opposite of the truth.** Our first attempt held  $V$  constant and compared raw accuracy. Every removal appeared to *improve* accuracy, because every removal escalated more, and at fixed  $V$  nothing prevents a variant from buying accuracy with latency. That is the error this paper spends §5.1 warning about, committed on ourselves. Each variant is therefore swept to a frontier and read at a fixed latency budget (Table 32).

The losses are real but wildly unequal, so the table ranks what matters rather than listing what helps. **At 400 ms the ladder is what makes the budget usable at all:** binary escalation reaches 0.281, which is always-cheap to within 0.003, because no binary policy can afford the top rung inside that budget. And per-example pricing costs  $-0.003$   $[-0.004, -0.001]$ : an interval excluding zero, so a

real effect, and small enough that we report it as negligible rather than as nothing. Those are different claims and the check that guards this table asserts the second.

## C A second serving model, and cross-model transfer

**Replication on a second serving model.** Every result above has one model deciding for itself, so we repeated the entire chain (oracle run, activation extraction, probe fitting, rate sweep) with SmolVLM-256M as the model that answers, on the same 1000 images and questions (Table 33).

The finding replicates and strengthens. On the smaller model the probe’s advantage on the joint target widens from  $+0.010$  to  $+0.066$ , and at the highest escalation rate both signals reach 0.455 accuracy while the probe costs 190.3 ms against 278.0 ms, 32% less, against 12 to 22% on the 500M.

Two properties survive the halving of model size and one does not. Self-knowledge is unchanged: AUROC on “will the cheap pass be correct” is 0.751 against 0.758, a difference of seven thousandths for a model with half the parameters and 16 fewer accuracy points. The escalation *opportunity* also grows, escalation repairs 24% of queries against 21%, and damages 3% against 4%. But knowing whether escalation will help degrades, from 0.734 to 0.672: half of the smaller model’s failures are unsolvable at any resolution, so the recoverability signal has less to work with. A weaker model needs routing more and can route slightly less well.

**Cross-model transfer.** Following the Encoder-Target Decoupling of [7], we predict the 500M model’s escalation outcomes from SmolVLM-256M activations on the same images. This scores 0.785  $[0.692, 0.873]$  at layer 30 and 0.770 at layer 1: at least as well as the 500M model predicts its own. The probe can therefore run off the serving path entirely, on a separate model that can be batched or placed on different hardware.

## D An adversary on the allocation signal

**The signal an attacker cannot see.** A routing rule is an attack surface. Liu et al. [15] force deferral in multimodal cascades with a border-confined trigger that flattens the weak model’s output distribution, inflating the provider’s bill without changing any accuracy metric. Their attack is metric-agnostic precisely because it targets that distribution, which the probe never reads. As a cheap proxy we applied unoptimised border perturbations covering 36% of the image, with the probe fitted on training examples only (Table 34).

A perturbation requiring no optimisation, no model access and no knowledge of the deployed metric inflates compute by 24% relative when allocation reads output entropy, and by nothing when it reads the probe. We claim only the weak form: an attacker optimising the output distribution is not optimising a projection onto a direction they cannot observe, and here that indirection suffices. A white-box attacker who knows the probe direction is the case this does not settle.

## E Abstention as a third action

We defined three targets in §3.3 and have used two.  $Q_3$  (answerable at all?) is negative for 30.6% of the pilot: no configuration we run answers those queries correctly. A system that could recognise them would decline instead of spending, which is the third action the efficiency literature omits and the selective-prediction literature is built around [18, 23]. Two axes decide whether it is worth adding, and they disagree.

**As a way to hold risk down, abstention is the weaker lever.** Under a risk constraint the alternative to declining is to spend more and re-decide. Sweeping both thresholds and charging escalated queries for both passes (Table 35), escalation reaches coverage that abstention cannot.

At a 50% tolerance escalation reaches full coverage while escalating 38% of traffic. The comparison needs a cost-instrumented multi-configuration run, which is the artefact the efficiency literature builds and the selective-prediction literature does not, and it says the two fields have been pulling different levers on the same signal.

**As a way to save compute, abstention is nearly worthless here, because the rule got there first.** Escalating a query no configuration can answer is pure waste, and 30.0% of the test fold is in that state. If the gain rule of §8 spent its escalation budget blind to solvability, a perfect abstention gate would recover that entire share. It does not (Table 36).

Calibrating on the signed gain already suppresses these queries: they have  $\mathbb{E}[G \mid x] \approx 0$  whatever their error probability, so the rule declines them for the same reason it declines confident failures. The residual headroom for a perfect, free abstention detector is 4.2% of latency at  $V = 800$ . **A third action is not where the remaining budget is.** What the escalation budget actually buys is visible in the same run: 54% of escalated queries have zero gain, escalation changes nothing, and that, not unanswerability, is the waste worth attacking.

**If a guarantee is needed rather than a preference.** Both §8’s rule and the thresholds above are calibrated to a point estimate. Where an operator needs a distribution-free promise instead, the three-way split of [13] transfers

directly, with its middle regime spending compute rather than widening a prediction set: at answer-level  $\alpha = 0.5$  the policy answers 52%, escalates 38% and declines 9%, and its coverage guarantee holds out of sample to within  $+0.025$ . A recall floor in the style of [12] gives the complementary guarantee (retain at least 90% of the queries escalation would have fixed, achieved 97% on the test fold) and cannot collapse to never escalating by construction, which is the failure mode our cost-minimising tuners exhibited. We report these as available machinery rather than as our headline policy, since neither improves on the cost-derived rule when a preference can be stated.

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| claim                                                        | status                                                                                                                                                                                                                                           |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The rate parameter is removable given a stated $V$ (Prop. 1) | demonstrated analytically, and it is a transposition of [11]                                                                                                                                                                                     |
| Entropy does not clear the randomisation hull                | measured on one mixture, both costings, all operating points                                                                                                                                                                                     |
| A free image descriptor matches the probe                    | measured on one mixture; untested on traffic where size and content decouple                                                                                                                                                                     |
| The two regimes require opposite signals                     | measured on eight corpus-model pairs. As a ranking claim it holds on all eight; as a policy claim the free descriptor clears on four and the model-read signal on five, of which one is withdrawn because a signal-free policy does better there |
| The probe’s pooled advantage is a domain effect              | measured three ways; the magnitude within a domain is uncertain                                                                                                                                                                                  |
| Accuracy saturates by 1152 px                                | established on DocVQA over four models including one at 0.892; not established on our second corpus, where the ceiling is above our rungs                                                                                                        |
| Pixels bind and tokens do not                                | measured on the one model whose two axes move separately; not decomposed for the other four                                                                                                                                                      |
| The ceiling bounds any model’s need from above               | observed on five models, one counterexample would refute it                                                                                                                                                                                      |
| Graded actions add value beyond the signal                   | measured on DocVQA only; on ChartQA the corpus supports one rung and the two rules coincide                                                                                                                                                      |
| Escalation value is domain-determined                        | observed across four datasets, assumed to extend to unseen ones                                                                                                                                                                                  |

Table 29: Evidential status of the paper’s claims. “Measured” means an interval was computed on held-out data; “observed” means the pattern is consistent across the runs we have without an interval that would detect its absence.

|                         | $2 \times 2$ ( $n = 500$ ) | $4 \times 4$ ( $n = 400$ ) |
|-------------------------|----------------------------|----------------------------|
| depths beating random   | 1 of 6                     | <b>4 of 4</b>              |
| best margin             | +0.006                     | +0.009 [0.005, 0.014]      |
| headroom to the oracle  | 0.122                      | 0.162                      |
| share of the gap closed | n/a                        | <b>6%</b>                  |

Table 30: Localizer margin over random cell choice, 60 resampled splits. Random is recomputed on each test fold rather than assumed, since the cell-correctness rate varies by fold.

| family            | configuration | tokens     | accuracy     |
|-------------------|---------------|------------|--------------|
| resolution        | 384 px        | 64         | 0.383        |
| resolution        | 768 px        | 320        | 0.516        |
| resolution        | full          | 320        | 0.549        |
| crop, random cell | $2 \times 2$  | <b>128</b> | 0.392        |
| crop, best cell   | $2 \times 2$  | <b>128</b> | <b>0.584</b> |

Table 31: Accuracy against visual tokens,  $n = 1000$ , two ways of spending the same budget. Token counts are medians: the 768 px and full configurations share one because the processor caps its target at the input’s longest side, and they differ on the 44% of images large enough for the cap not to bind (§10). Paired against the best resolution configuration, the best crop is +0.035 [0.008, 0.063] more accurate at  $2.5\times$  fewer tokens.

| variant                     | acc. at 400 ms | acc. at 500 ms |
|-----------------------------|----------------|----------------|
| <b>reference</b>            | <b>0.653</b>   | <b>0.743</b>   |
| no per-example pricing      | 0.650 (−0.003) | 0.744 (+0.000) |
| no calibration (tuned rate) | 0.431 (−0.221) | 0.571 (−0.172) |
| no ladder (binary)          | 0.281 (−0.372) | 0.622 (−0.121) |
| no gain target (UCCI)       | 0.278 (−0.375) | 0.611 (−0.133) |
| no signal (fixed policy)    | 0.278 (−0.375) | 0.278 (−0.465) |

Table 32: Each variant swept over six operator preferences and read at a fixed budget, 1200 DocVQA pages, 30 resamples. Deltas are paired against the reference frontier.

| serving model | cheap acc. | full acc. | probe AUROC | entropy AUROC |
|---------------|------------|-----------|-------------|---------------|
| SmolVLM-500M  | 0.383      | 0.549     | 0.761       | 0.751         |
| SmolVLM-256M  | 0.298      | 0.511     | 0.682       | 0.616         |

Table 33: AUROC is on the joint escalation-gain target. On the reported fold of both models every entropy operating point is Pareto-dominated; six probe points lie on the front for the 256M model, four for the 500M. The resplit test of Fig. 8 was run for the 500M model only.

| perturbation         | signal  | shift          | escalation                   |
|----------------------|---------|----------------|------------------------------|
| high-frequency noise | entropy | +0.28 $\sigma$ | 50% $\rightarrow$ <b>62%</b> |
| high-frequency noise | probe   | −0.04 $\sigma$ | 50% $\rightarrow$ 50%        |
| flat fill            | entropy | +0.19 $\sigma$ | 50% $\rightarrow$ <b>56%</b> |
| flat fill            | probe   | +0.01 $\sigma$ | 50% $\rightarrow$ 50%        |

Table 34: Border perturbations touching no central pixel, 80 held-out images. Two structurally opposite perturbations both raise output entropy and neither moves the probe.

| risk tolerance | abstain only | or escalate | cost  |
|----------------|--------------|-------------|-------|
| $\leq 30\%$    | unreachable  | <b>58%</b>  | 320.8 |
| $\leq 40\%$    | 41%          | <b>88%</b>  | 312.2 |
| $\leq 50\%$    | 69%          | <b>100%</b> | 200.7 |

Table 35: Maximum coverage at each risk tolerance, latency in milliseconds. At a 30% tolerance abstention alone serves no query at acceptable risk.

| $V$  | escalates | of which hopeless | perfect gate saves |
|------|-----------|-------------------|--------------------|
| 400  | 24%       | 17%               | 2.8%               |
| 800  | 34%       | 19%               | 4.2%               |
| 1600 | 40%       | 22%               | 5.3%               |

Table 36: Escalation is aimed below the 30.0% base rate of unsolvable queries, so an oracle abstention gate (which cannot lose accuracy, since those queries are wrong under every action) recovers only the margin.